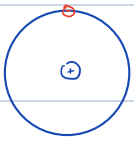
Quantum Mechanics

$$I(\lambda, T) = \frac{2hc^2}{\lambda^5} \frac{1}{e^{\left(\frac{hc}{\lambda kT}\right)} - 1}$$

$E = hf = \hbar\omega$. $p = \frac{h}{\lambda}$ $\quad n\lambda_n = 2\pi r_n$



$$\frac{mv^2}{r} = \frac{zq^2}{4\pi\varepsilon_0 r^2}$$

$$\Rightarrow v_n = \frac{zq^2}{4\pi\varepsilon_0 n\hbar} \quad r_n = \frac{n\hbar}{mv_n} = \frac{(n\hbar)^2 4\pi\varepsilon_0}{zq^2 m}$$

$$E = T + V = \frac{1}{2}mv^2 + \left(-\frac{zq^2}{4\pi\varepsilon_0 r}\right) = -\frac{1}{2}mv^2 = -\frac{1}{2}m\left(\frac{zq^2}{4\pi\varepsilon_0\hbar}\right)^2 \frac{1}{n^2} \xrightarrow{z=1} -\frac{13.6\,eV}{n^2}$$

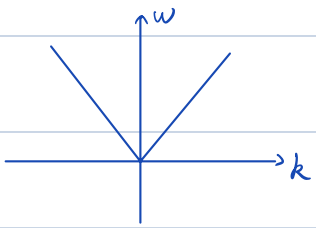
Schrödinger equation

平面波 $\Rightarrow y(x,t) = y_0 e^{i(kx-\omega t)} \,/\, y_0 e^{i(-kx-\omega t)}$

$$\frac{\partial^2 y}{\partial t^2} = v^2\left(\frac{\partial^2 y}{\partial x^2}\right)$$

$$\Rightarrow (-i\omega)^2 y = v^2 (ik)^2 y$$

$$\omega = \pm vk$$



$$\begin{cases} E = \hbar\omega \\ p = \frac{h}{\lambda} = \frac{h}{\frac{2\pi}{k}} = \hbar k \end{cases} \quad \text{波数} \quad \Rightarrow E = \frac{p^2}{2m} + V(x) \Rightarrow \hbar\omega = \frac{\hbar^2 k^2}{2m} + V(x)$$

$\psi(x,t)$ $\quad \boxed{i\hbar\frac{\partial\psi}{\partial t} = -\frac{\hbar^2}{2m}\frac{\partial^2\psi}{\partial x^2} + V\cdot\psi}$ 1D

$$\hbar\omega = \frac{\hbar^2(k_x^2 + k_y^2 + k_z^2)}{2m} + V(x) \quad .$$

$$\boxed{i\hbar\frac{\partial\psi}{\partial t} = -\frac{\hbar^2}{2m}\left[\frac{\partial^2\psi}{\partial x^2} + \frac{\partial^2\psi}{\partial y^2} + \frac{\partial^2\psi}{\partial z^2}\right] + V\psi} \quad 3D$$

ψ : $|\psi|^2 dV$: probability of finding the particle in dV

$\Rightarrow |\psi|^2$ is real . $|\psi|^2 = \psi^* \cdot \psi$

$$\int_{-\infty}^{+\infty} |\psi|^2 dV = 1$$

Expectation of position : $\langle x\rangle = \int x\cdot|\psi(x,t)|^2 dx = \int_{-\infty}^{+\infty} \psi^*(x,t)\cdot x\cdot\psi(x,t)\,dx$

$$\langle x^2\rangle = \int_{-\infty}^{+\infty} \psi^*(x,t)\cdot x^2\cdot\psi(x,t)$$

$$\Delta x^2 = \langle x^2\rangle - \langle x\rangle^2$$

Assume $\psi(x,t) = \varphi(x)\cdot a(t)$

$$i\hbar\frac{a'(t)}{a(t)} = -\frac{\hbar^2}{2m}\frac{\varphi''(x)}{\varphi(x)} + V(x) = E$$

$$\Rightarrow \begin{cases} i\hbar\frac{a'(t)}{a(t)} = E \Rightarrow a'(t) = \frac{E}{i\hbar}a(t), \quad a(t) = e^{i\frac{E}{\hbar}t} \\ -\frac{\hbar^2}{2m}\varphi'' + V(x)\varphi = E\varphi \end{cases}$$

consider $V(x) = 0$, $k^2 = \frac{2mE}{\hbar^2}$

$$\boxed{\varphi''(x) + k^2\varphi(x) = 0}$$

$$\Rightarrow \varphi(x) = \frac{1}{\sqrt{L}}e^{ikx} \qquad L\to+\infty \qquad \int_{-\frac{L}{2}}^{\frac{L}{2}}|\varphi(x)|^2dx = 1$$

$$\varphi(x) = \frac{1}{\sqrt{L}}e^{-ikx} \qquad |\varphi(x)| = \text{Constant}$$

$p = \hbar k$ $\quad$ $p = \hbar(-k)$

$\hat{p}$: operator $\quad \boxed{\hat{p}\cdot\varphi(x) = p\cdot\varphi(x)}$

$$\hat{p}\cdot e^{ikx} = \hbar k\cdot e^{ikx} \qquad \hat{p}: -i\hbar\frac{\partial}{\partial x}$$

$$\langle p\rangle = \int\varphi^*\cdot\hat{p}\cdot\varphi\,dx = \int\varphi^*\cdot(-i\hbar\frac{\partial}{\partial x})\cdot\varphi\,dx$$

$$\langle x\rangle = \int\varphi^*\cdot\hat{x}\cdot\varphi\,dx = \int\varphi\cdot x\cdot\varphi\,dx$$

$$\hat{x}\cdot\hat{p}\cdot\varphi(x) = x\cdot(-i\hbar\frac{\partial}{\partial x}\varphi(x))$$

$$\hat{p}\cdot\hat{x}\cdot\varphi(x) = -i\hbar\frac{\partial}{\partial x}(x\cdot\varphi(x)) = -i\hbar\varphi(x) + x\cdot(-i\hbar\frac{\partial}{\partial x})\varphi(x)$$

$$\Rightarrow \hat{x}\hat{p}\cdot\varphi(x) - \hat{p}\cdot\hat{x}\cdot\varphi(x) = i\hbar\varphi(x)$$

$$(\hat{x}\hat{p} - \hat{p}\hat{x}) = i\hbar$$

$$[\hat{x},\hat{p}] = \hat{x}\hat{p} - \hat{p}\hat{x} = i\hbar$$

$$\langle\hat{p}\rangle = \int\varphi^*\hat{p}\cdot\varphi\,dx = \int\varphi^*(-i\hbar\frac{\partial}{\partial x})\cdot\varphi\,dx$$

$$\langle\hat{p}^2\rangle = \int\varphi^*\hat{p}^2\cdot\varphi\,dx = \int\varphi^*(-i\hbar\frac{\partial}{\partial x})^2\varphi\,dx = (-i\hbar)^2\int\varphi^*\frac{\partial^2}{\partial x^2}\varphi\,dx$$

$$\Delta p^2 = \langle p^2\rangle - \langle p\rangle^2$$

$$\Delta x^2 = \langle x^2\rangle - \langle x\rangle^2$$

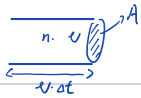
$\Delta p\cdot\Delta x \geq \frac{\hbar}{2}$ 海森堡不确定性原理

$\Delta E\cdot\Delta t \geq \frac{\hbar}{2}$

$\vec{j}$ 流密度

经典物理

$\frac{\vec{p}}{m} = \vec{v}$. $j = n\vec{v}$.



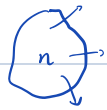
$j = \frac{n \cdot v \cdot dt \cdot A}{dt \cdot A}$. $\vec{j} = n\vec{v}$

量子力学

$n = |\psi|^2$. $\vec{j} \sim |\psi|^2 \vec{v} \sim |\psi|^2 \frac{\vec{p}}{m} \sim Re\{\psi^* \hat{\frac{\vec{p}}{m}} \psi\}$

$\vec{j} = Re\{\psi^* \hat{\frac{\vec{p}}{m}} \psi\} = \frac{1}{2m}[\psi^*(-i\hbar\nabla)\psi + \psi(i\hbar\nabla)\psi^*]$ ··· 3D

连续性方程: $\int \frac{\partial n}{\partial t} dV = -\oint \vec{j} \cdot d\vec{s} = -\int \nabla \cdot \vec{j} \, dV$



$\Rightarrow \frac{\partial n}{\partial t} = -\nabla \vec{j}$

$n = \psi^* \psi$. $\frac{\partial n}{\partial t} = \frac{\partial \psi^*}{\partial t} \psi + \frac{\partial \psi}{\partial t} \psi^*$

$$\begin{cases} \psi^* \cdot i\hbar \frac{\partial \psi}{\partial t} = -\frac{\hbar^2}{2m} \psi^* \nabla^2 \psi + \psi^* V \psi \\ \psi \left(-i\hbar \frac{\partial \psi^*}{\partial t}\right) = -\frac{\hbar^2}{2m} \psi \nabla^2 \psi^* + \psi V \psi^* \end{cases}$$

$\Rightarrow i\hbar \underbrace{[\psi^* \frac{\partial \psi}{\partial t} + \psi \frac{\partial \psi^*}{\partial t}]}_{\frac{\partial n}{\partial t}} = -\frac{\hbar^2}{2m}[\psi^* \nabla^2 \psi - \psi \nabla^2 \psi^*]$

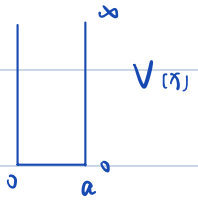
$\Rightarrow \frac{\partial n}{\partial t} = -[-i\frac{\hbar}{m}][\psi^* \nabla^2 \psi - \psi \nabla^2 \psi^*]$

$= -\nabla \underbrace{[-i\frac{\hbar}{2m}(\psi^* \nabla \psi - \psi \nabla \psi^*)]}_{\vec{j}}$

$\Rightarrow \vec{j} = -i\frac{\hbar}{2m}(\psi^* \nabla \psi - \psi \nabla \psi^*)$

$$-\frac{\hbar^2}{2m}\varphi'' + V(x)\cdot\varphi = E\varphi$$

eg 1. $V(x) = \begin{cases} 0, & 0<x<a \\ \infty, & \text{otherwise} \end{cases}$



① $x \le 0,\ x \ge a,\quad \varphi(x) = 0$

② $0 < x < a,\quad -\frac{\hbar^2}{2m}\cdot\varphi'' = E\cdot\varphi$

$E > 0,\quad k^2 = \frac{2mE}{\hbar^2} > 0,\quad \varphi'' + k^2\varphi = 0$

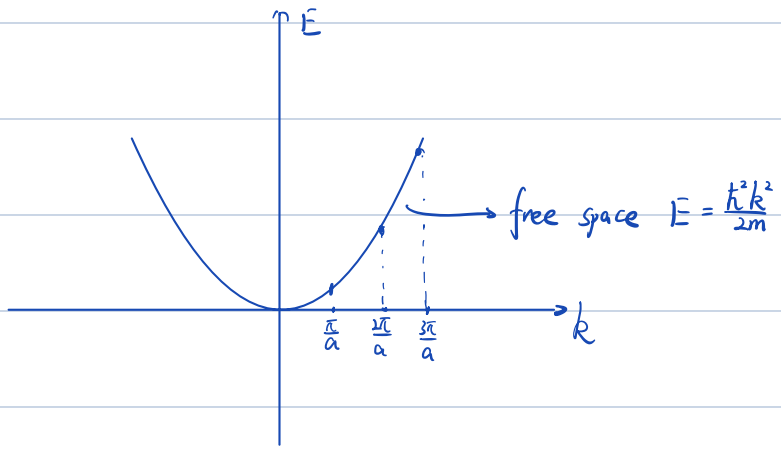
$$\Rightarrow \varphi(x) = A\cdot e^{ikx} + Be^{-ikx} = C\cdot\sin kx + D\cdot\cos kx$$

$$\begin{cases} \varphi(0^+) = \varphi(0^-) \Rightarrow D = 0 \\ \varphi(a^+) = \varphi(a^-) \Rightarrow C = 0 \ / \ \sin ka = 0 \end{cases}$$

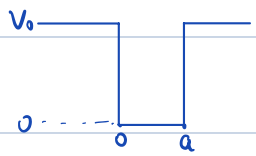
$$\Rightarrow \begin{cases} D = 0 \\ ka = n\pi \end{cases} \quad E_n = \frac{\hbar^2 k^2}{2m} = \frac{\hbar^2}{2m}\left(\frac{n\pi}{a}\right)^2$$

$$\Rightarrow \varphi_n(x) = \begin{cases} 0, & x<0,\ x>a \\ C\cdot\sin\left(\frac{n\pi}{a}x\right), & 0 \le x \le a \end{cases}$$

$$\int_{-\infty}^{+\infty} |\varphi_n(x)|^2 dx = 1 \longrightarrow \varphi_n(x) = \begin{cases} 0, & x<0,\ x>a \\ \sqrt{\frac{2}{a}}\cdot\sin\left(\frac{n\pi}{a}x\right), & 0 \le x \le a \end{cases}$$



eg 2. $V(x) = \begin{cases} 0, & 0<x<a \\ V_0 & \text{otherwise} \end{cases}$



$0<x<a$: $\varphi(x) = A\sin kx + B\cos kx$

$x>a,\ x<0$: $-\frac{\hbar^2}{2m}\varphi'' + V_0\varphi = E\varphi$. $-\frac{\hbar^2}{2m}\varphi'' = (E-V_0)\varphi$ $\quad E<V_0$

$\varphi'' = \frac{2m}{\hbar^2}(V_0 - E)\varphi$

$\xrightarrow{\alpha^2 = \frac{2m}{\hbar^2}(V_0-E)} \varphi'' = \alpha^2\varphi$

$x<0$. $\varphi(x) = Ce^{\alpha x} + Me^{-\alpha x}$

$x>a$. $\varphi(x) = Ne^{\alpha x} + De^{-\alpha x}$

束缚态 : $\varphi(+\infty) \longrightarrow 0$. $\Rightarrow M=0,\ N=0$

$\begin{cases} \varphi(x) \text{ 连续} \\ \text{if } |V(x)|<\infty \quad \varphi'(x) \text{ 连续} \end{cases}$

$$\Rightarrow \begin{cases} C = B \\ \alpha C = Ak \\ De^{-\alpha a} = A\sin ka + B\cos ka \\ -\alpha De^{-\alpha a} = Ak\cos ka - Bk\sin ka \end{cases} \qquad \begin{bmatrix} 0 & 1 & -1 & 0 \\ -k & 0 & \alpha & 0 \\ \sin ka & \cos ka & 0 & -e^{-\alpha a} \\ k\cos ka & -k\sin ka & 0 & \alpha e^{-\alpha a} \end{bmatrix} \begin{bmatrix} A \\ B \\ C \\ D \end{bmatrix} = 0$$

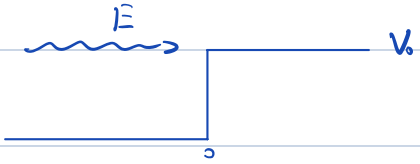
$\hat{M}(E)\begin{pmatrix} A \\ B \\ C \\ D \end{pmatrix} = 0 \Rightarrow \det(\hat{M}(E)) = 0$

$\Rightarrow \tan(ka) = \frac{2\alpha k}{k^2-\alpha^2}$. k, α is function of E

$\xrightarrow{\xi = \frac{E}{V_0},\ \alpha_0 = \sqrt{\frac{2mV_0}{\hbar^2}}} \tan(\alpha_0 a\sqrt{\xi}) = \frac{2\sqrt{\xi(1-\xi)}}{2\xi-1}$ at least one solution

$E = \xi^* V_0$

$\Rightarrow \begin{bmatrix} B \\ C \\ D \end{bmatrix} = \begin{pmatrix} f_1(A) \\ f_2(A) \\ f_3(A) \end{pmatrix}$ $\quad \int_{-\infty}^{+\infty} |\varphi|^2 dx = 1 \Rightarrow A$



$$V(x) = \begin{cases} 0 & x<0 \\ V_0 & x>0 \end{cases}$$

if $E > V_0$

$$\varphi(x) = \begin{cases} e^{ikx} + \underbrace{re^{-ikx}}_{\text{反射}} & x<0 \\ \underbrace{t\cdot e^{ik'x}}_{\text{透射}} \end{cases}$$

$-\frac{\hbar^2}{2m}\varphi'' + V(x)\varphi = E\varphi$，代入 $\varphi(x)$

$x<0$ $\quad E = \frac{\hbar^2 k^2}{2m}$，$k = \frac{\sqrt{2mE}}{\hbar}$

$x>0$ $\quad E = \frac{\hbar^2 k'^2}{2m} + V_0$，$k' = \frac{\sqrt{2m(E-V_0)}}{\hbar}$

$\varphi(0^-) = \varphi(0^+) \Rightarrow 1+r = t$

$\varphi'(0^-) = \varphi'(0^+) \Rightarrow ik(1-r) = ik't \Rightarrow 1-r = \frac{k'}{k}t$

$\Rightarrow t = \frac{2k}{k+k'}$，$r = \frac{k-k'}{k+k'}$

反射率 $R = \left|\frac{j_R}{j_{in}}\right|$ $\quad j_{in} = Re\{\psi_{in}^* \cdot \frac{\hat{p}}{m}\cdot \psi_{in}\} \xrightarrow{\psi_{in} = e^{ikx}} Re\{e^{ikx}\frac{-i\hbar\frac{d}{dx}}{m}e^{-ikx}\} = \frac{\hbar k}{m}$

$j_r = Re\{\psi_{ref}^* \cdot \frac{\hat{p}}{m}\cdot \psi_{ref}\} \xrightarrow{\psi_{ref} = re^{-ikx}} Re\{re^{ikx}\frac{-i\hbar\frac{d}{dx}}{m}re^{-ikx}\} = \frac{\hbar k}{m}\cdot|r^2|$

$\Rightarrow R = |r^2|$

透射率 $T = \left|\frac{j_T}{j_{in}}\right|$

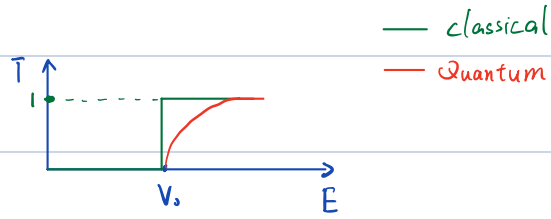
$j_T = Re\{\psi_T^* \cdot \frac{\hat{p}}{m}\cdot \psi_T\} = Re\{\frac{\hbar k'}{m}\}|t|^2$

$\Rightarrow T = Re\{\frac{k'}{k}\}|t|^2$

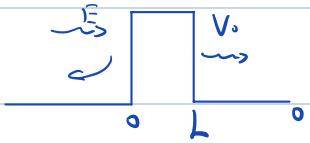
If $E < V_0$. $\quad k' = \frac{\sqrt{2m(E-V_0)}}{\hbar} = i\cdot k$. $\quad k = \sqrt{2m(V_0-E)}/\hbar$

$r = \frac{k-k'}{k+k'} = \frac{k-ik}{k+ik}$ $\quad t = \frac{2k}{k+k'} = \frac{2k}{k+ik}$

$R = |r|^2 = \left|\frac{k-ik}{k+ik}\right|^2 = 1$ $\quad T = Re\{\frac{k'}{k}\}\cdot|k|^2 = Re\{\frac{ik}{k}\}\cdot|t'| = 0$ $\quad \boxed{R+T=1}$



$$V(x) = \begin{cases} 0 & x<0,\ x>L \\ V_0 & L>x>0 \end{cases}$$



$$\varphi(x) = \begin{cases} e^{ikx} + re^{-ikx} & x<0 \\ Ae^{ik'x} + Be^{-ik'x} & 0<x<L \\ te^{ikx} & x>L \end{cases}$$

$k = \sqrt{2mE}/\hbar$. $k' = \sqrt{2m(E-V_0)}/\hbar$.

$E > V_0$. k' is real

$E < V_0$. $k' = ik$. $k = \frac{\sqrt{2m(V_0-E)}}{\hbar}$

$$\begin{cases} \varphi(0^-) = \varphi(0^+) \\ \varphi'(0^-) = \varphi'(0^+) \\ \varphi(L^-) = \varphi(L^+) \\ \varphi'(L^-) = \varphi'(L^+) \end{cases} \Rightarrow T = |t|^2 = \frac{4k^2k'^2}{[(k^2+k'^2)^2\sin^2k'L + 4k^2k'^2\cos^2k'L]}$$

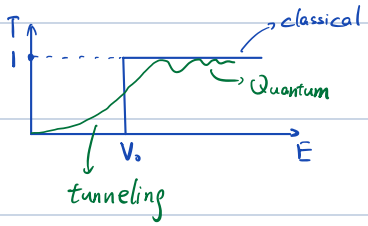
$\cos k'L = 1$, $\sin k'L = 0$. T maximum. $k'L = 0, \pi, 2\pi \dots$ $k' = \frac{2\pi}{\lambda}$.

$\Rightarrow L = 0, \frac{\lambda}{2}, \lambda, \dots \frac{n\pi}{k'}$. oscillation

if $E < V_0$. $k' = ik$

$$T = \frac{4k^2k^2}{[(k^2-k^2)\sinh^2kL + 4k^2k^2\cosh^2kL]}$$

$E < V_0$. $T \neq 0$ $\Rightarrow$ tunneling. 隧穿



Band Structure of Semiconductors

)⊕( )⊕( )⊕( )⊕(. $V(x+p) = V(x)$. 周期性

$V(\vec{r} + m_1\vec{a} + m_2\vec{b} + m_3\vec{c}) = V(\vec{r})$

$|\varphi(x+p)|^2 = |\varphi(x)|^2$ $\varphi(x+p) = \varphi(x)\cdot e^{ikp}$. k is real

Bloch Theorem

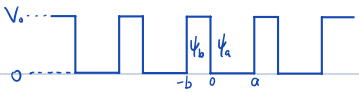
$\varphi(x) = U_k(x)e^{ikx}$

$\varphi(x+p) = U_k(x+p)\,e^{jk(x+p)} = U_k(x)\cdot e^{ikx}\cdot e^{ikp}$

$\Rightarrow U_k(x+p) = U_k(x)$ ⟶ Bloch function.

在周期性势中 波函数 : $\varphi(x) = U_k(x)\cdot e^{ikx}$ free space : $\varphi(x) = e^{ikx}$

Krony - Penney model



$0 \le x \le a$.

$-\frac{\hbar^2}{2m}\cdot\varphi_a'' = E\cdot\varphi_a$. $\varphi_a'' + \frac{2mE}{\hbar^2}\cdot\varphi_a = 0$. $\alpha^2 = \frac{2mE}{\hbar^2}$

$\Rightarrow \varphi_a(x) = A_a\sin\alpha x + B_a\cos\alpha x$.

$-b \le x \le 0$.

$-\frac{\hbar^2}{2m}\cdot\varphi_b'' + V_0\cdot\varphi_b = E\cdot\varphi_b$. $\varphi_b'' + \frac{2m(E-V_0)}{\hbar^2}\cdot\varphi_b = 0$. $\beta = \frac{\sqrt{2m(E-V_0)}}{\hbar}$.

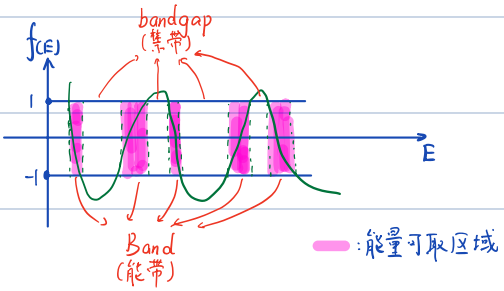
$\Rightarrow \varphi_b(x) = A_b\sin\beta x + B_b\cos\beta x$

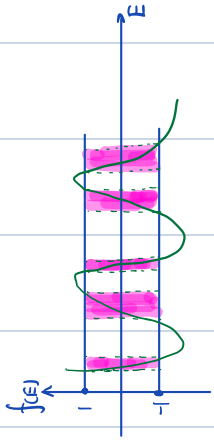
$$\begin{cases} \varphi_a(0) = \varphi_b(0) \\ \varphi_a'(0) = \varphi_b'(0) \\ \varphi_a(a) = \varphi_b(-b)\cdot e^{ikp} \\ \varphi_a'(a) = \varphi_b'(-b)\cdot e^{ikp} \end{cases} \Rightarrow \text{Bloch Theorem} \Rightarrow \begin{cases} B_a = B_b \\ A_a\cdot\alpha = A_b\cdot\beta \\ A_a\sin\alpha a + B_a\cos\alpha a = (-A_b\sin\beta b + B_b\cos\beta b)\cdot e^{ikp} \\ A_a\alpha\cdot\cos\alpha a - B_a\alpha\cdot\sin\alpha a = (A_b\beta\cdot\cos\beta b + B_b\beta\cdot\sin\beta b)\,e^{ikp} \end{cases}$$

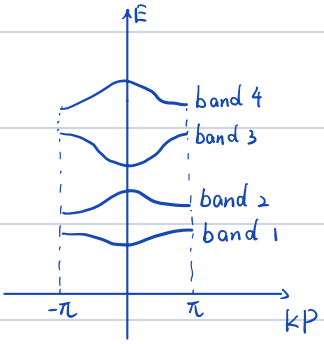
$$[\tilde{M}(E,k)]\cdot\begin{bmatrix} A_a \\ A_b \\ B_a \\ B_b \end{bmatrix} = 0 \quad . \quad \det(\tilde{M}(E,k)) = 0 \Rightarrow \cos kp = -\frac{(\alpha^2+\beta^2)}{2\alpha\beta}(\sin\alpha a)\cdot(\sin\beta b) + (\cos\alpha a)(\cos\beta b) \longrightarrow f(E)$$

$\alpha = \frac{\sqrt{2mE}}{\hbar}$. $\beta = \frac{\sqrt{2m(E-V_0)}}{\hbar}$. a function of E.

k real . $-1 \le \cos kp \le 1$. $-1 \le f(E) \le 1$







$\omega s kP = f(E)$

1 2 3 4 ... N

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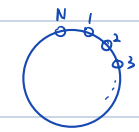
$\varphi(x+p) = \varphi(x) e^{ikp}$

$\varphi(x+2p) = \varphi(x) \cdot e^{2ikp}$

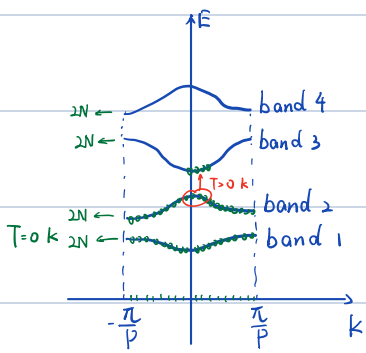


periodic boundary condition

$\varphi(x+Np) = \varphi(x)$

$\Rightarrow \varphi(x+Np) = \varphi(x) e^{Nikp} = \varphi(x)$

$\Rightarrow Nkp = 2\pi n \cdot \quad k_n = \frac{2\pi n}{Np} \quad \cdot \quad n \in (-\frac{N}{2}, \frac{N}{2}]$



$\Delta k = \frac{2\pi}{Np}$

相速度: v_p

群速度: $v_g = \frac{\partial \omega}{\partial k} = \frac{1}{\hbar} \frac{\partial E}{\partial k}$

velocity of electron

$\Delta E \cdot \Delta t \geq \frac{\hbar}{2}$

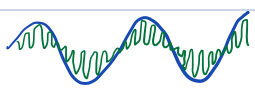
$\Delta p \cdot \Delta x \geq \frac{\hbar}{2}$

$\varphi \sim A e^{j(kx-\omega t)} + B e^{j((k+\Delta k)x-(\omega+\Delta\omega)t)}$

$= e^{j(kx-\omega t)} \cdot [A + B e^{j(\Delta k x - \Delta\omega t)}]$

$\Delta k \ll k$

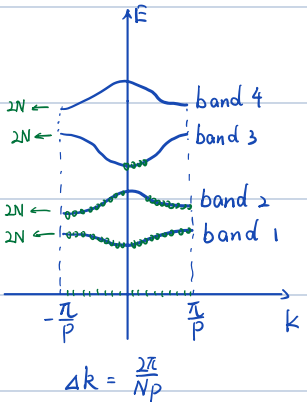
$\lambda = \frac{2\pi}{k} \gg \lambda' = \frac{2\pi}{\Delta k}$



$\Delta k x - \Delta\omega t = \text{constant}$

$\Delta k \cdot \frac{dx}{dt} - \Delta\omega = 0$

$v_g = \frac{dx}{dt} = \frac{\Delta\omega}{\Delta k} = \frac{\partial \omega}{\partial k} \xrightarrow{E = \hbar \cdot \omega} \frac{1}{\hbar} \cdot \frac{\partial E}{\partial k}$



Band 1 : filled Band

$J \sim q \cdot \Sigma v = 0$

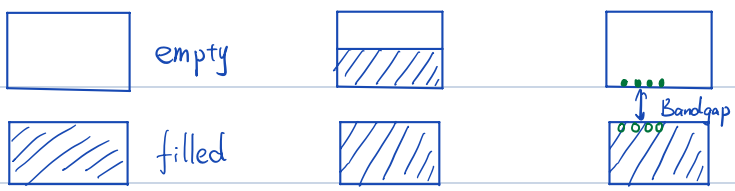
Band 4 : empty Band

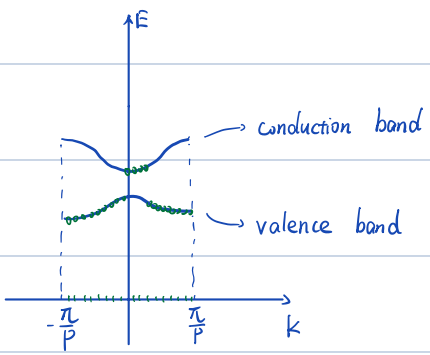
$J \sim q \cdot \Sigma v = 0$

Band 2 & 3. partially filled Band

$J \sim q \cdot \Sigma v \neq 0$

insulators metal semiconductor





$v_g = \frac{1}{\hbar} \cdot \frac{\partial E}{\partial k}$. $dE = F \cdot dx = F \cdot v_g \cdot dt$

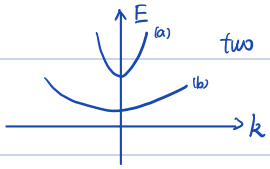
$$\begin{cases} \frac{dE}{dt} = F \cdot v_g = F \cdot \frac{1}{\hbar} \frac{\partial E}{\partial k} \\ \frac{dE}{dt} = \frac{\partial E}{\partial k} \cdot \frac{\partial k}{\partial t} \end{cases}$$

$$\Rightarrow \vec{F} = \hbar \cdot \frac{d\vec{k}}{dt}$$

$$a = \frac{dv_g}{dt} = \frac{1}{\hbar}\cdot\frac{d}{dt}\left(\frac{\partial E}{\partial k}\right) = \frac{1}{\hbar}\cdot\frac{\partial}{\partial k}\left(\frac{\partial E}{\partial k}\right)\cdot\frac{\partial k}{\partial t}$$

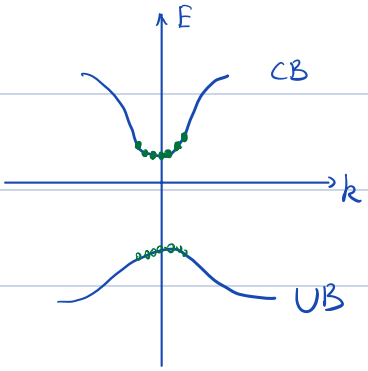
$$a = \frac{1}{\hbar}\cdot\frac{\partial^2 E}{\partial k^2}\cdot F$$

$$a = \frac{F}{m^*} \quad define: m^* = \frac{\hbar^2}{\frac{\partial^2 E}{\partial k^2}} \text{ 有效质量}$$



two metarials

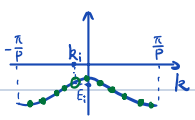
$m^*_{(a)} < m^*_{(b)}$



CB: $m^* > 0$

VB: $m^* < 0$

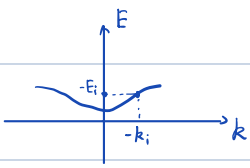
Hole



missing an electron at (k_i, E_i)

$$k_{total} = \sum_j k_j - k_i = -k_i \quad (\sum_j k_j = 0)$$

$$E_{total} = \sum_j E_j - E_i = \text{Constant} - E_i$$



exit a hole at $(-k_i, -E_i)$

① $v_g^{(h)} = \frac{1}{\hbar}\cdot\frac{\partial(-E_i)}{\partial(-k_i)} = \frac{1}{\hbar}\cdot\frac{\partial E_i}{\partial k_i} = v_g^{(E)}$

② $m^*_{(h)} = \frac{\hbar^2}{\frac{\partial^2(-E_i)}{\partial(-k_i)^2}} = \frac{\hbar^2}{-\frac{\partial^2 E_i}{\partial k_i^2}} = -m^*_{(E)}$

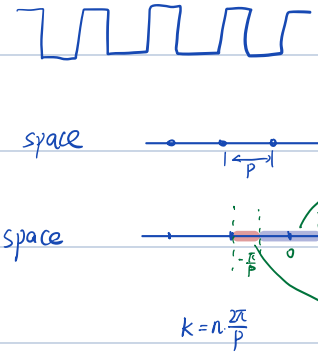
③ $F = \hbar\cdot\frac{d(k_{(h)})}{dt} = -\hbar\cdot\frac{dk_{(h)}}{dt} = -\hbar\frac{dk_i}{dt}$

$$\hbar\frac{dk_i}{dt} = -|e|\cdot E$$

$$\Rightarrow F = -(-|e|E) = |e|E$$

Brillouin Zone

1D.



3D. $\vec{R} = n_1\vec{a} + n_2\vec{b} + n_3\vec{c}$

FCC: $\vec{a} = \frac{a}{2}(0\,1\,1)$ $\vec{b} = \frac{a}{2}(1\,0\,1)$ $\vec{c} = \frac{a}{2}(1\,1\,0)$

k-space: $\vec{K} = n_1\vec{k}_1 + n_2\vec{k}_2 + n_3\vec{k}_3$

$\vec{k}_x \cdot \vec{a} = 2\pi$ $\vec{k}_1 \cdot \vec{b} = 0$
$\vec{k}_y \cdot \vec{b} = 2\pi$ ⋮
$\vec{k}_z \cdot \vec{c} = 2\pi$ $\vec{k}_x \cdot \vec{c} = 0$

$\Rightarrow k_x = 2\pi \cdot \frac{b \times c}{|a \cdot b \times c|}$ $k_y = 2\pi \cdot \frac{c \times a}{|a \cdot b \times c|}$ $k_z = 2\pi \cdot \frac{a \times b}{|a \cdot b \times c|}$

$k_x = 2\pi \cdot \frac{\frac{a^2}{4}(-1,1,1)}{\frac{a^3}{8} \times 2} = \frac{2\pi}{a}(-1,1,1)$

$k_y = \frac{2\pi}{a}(1,-1,1)$

$k_z = \frac{2\pi}{a}(1,1,-1)$

→ BCC

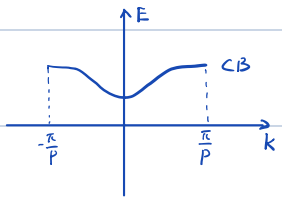
FCC ⟶ BCC

↓ ↓

real ⟶ k-space

Carrier Density.

of state $f(E)$



N k $\Rightarrow 2N$ state

$$N = \sum_i D(E_i) f(E_i) \approx \int \underbrace{D(E)}_{\text{态密度}} f(E)\, dE$$

# of state per E x probability of being filled

$D(E)\cdot dE$: # of state with energy between E and $E+dE$

$g(E) = \frac{D(E)}{V}$ ⟶ (归化)态密度

$E = \frac{\hbar^2}{2m}(k_x^2 + k_y^2 + k_z^2)$. $\varphi \sim e^{ik_x x} e^{ik_y y} e^{ik_z z}$

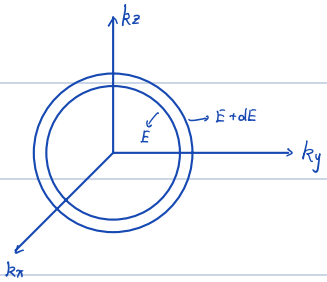
peroidic boundary condition : $\varphi(x,y,z) = \varphi(x+a,y,z) = \varphi(x,y+b,z) = \varphi(x,y,z+c)$

$\Rightarrow k_x = \frac{2\pi n_1}{a}$. $k_y = \frac{2\pi n_2}{b}$. $k_z = \frac{2\pi n_3}{c}$

$\Delta V_k = \Delta k_x \cdot \Delta k_y \cdot \Delta k_z = \frac{(2\pi)^3}{abc}$

每个k在k-space所占的体积

$E = \frac{\hbar^2}{2m}(k_x^2 + k_y^2 + k_z^2) = \frac{\hbar^2}{2m}k^2$

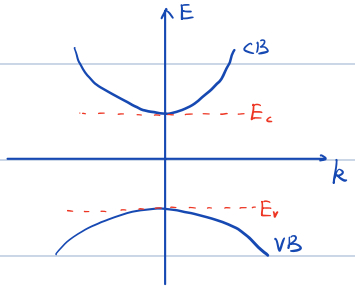


$D(E)\,dE = \frac{4\pi k^2 dk}{\Delta V_k} \times 2$ ⟶ 每个态有2个电子

$D(E) = abc \cdot \frac{k^2}{\pi^2}\cdot\frac{dk}{dE} = V\cdot\frac{k^2}{\pi^2}\cdot\frac{dk}{dE}$. $g(E) = \frac{D(E)}{V} = \frac{k^2}{\pi^2}\cdot\frac{dk}{dE}$

$k = \frac{\sqrt{2mE}}{\hbar} \Rightarrow g(E) = \frac{2mE}{\hbar^2}\cdot\frac{1}{\pi^2}\cdot\frac{1}{\hbar}\sqrt{\frac{m}{2}}\cdot\frac{1}{\sqrt{E}} = \frac{m\cdot\sqrt{2m}\cdot\sqrt{E}}{\pi^2\cdot\hbar^3}$

$$E = \frac{\hbar^2 k^2}{2m}$$



$$CB: E = E_c + \frac{\hbar^2 k^2}{2m_n^*}$$

$$VB: E = E_v - \frac{\hbar^2 k^2}{2m_p^*}$$

$$CB: g_{c(E)} = \frac{m_n^* \sqrt{2m_n^*} \cdot \sqrt{E - E_c}}{\pi^2 \hbar^3}, \quad E > E_c$$

$$VB: g_{v(E)} = \frac{m_p^* \sqrt{2m_p^*} \sqrt{E_v - E}}{\pi^2 \hbar^3}, \quad E < E_v$$

m_n^*, m_p^* ⇒ density of state effective mass

電子濃度: $n = \int_{E_c}^{E_c^{top}} g_{c(E)} \cdot f_{(E)} \, dE$

空穴濃度: $p = \int_{E_v^{bottom}}^{E_v} g_{v(E)} \cdot (1 - f_{(E)}) \, dE$

a system with fixed number of electrons and fixed total energy.

i) $N_t = \sum_i N_i$

ii) $E_t = \sum_i N_i E_i$

iii) pauli-principle : only one electron per state.

$N_t = 5$. $E_t = 12$

4 = E_3 □□□□□□□

2 = E_2 □□□□

0 = E_1 □□

4 = E_3 0 0 0 □□□□

2 = E_2 □□□□

0 = E_1 0 0 □

$C_2^2 \cdot C_4^0 \cdot C_8^3 = 56$

4 = E_3 0 0 □□□□

2 = E_2 0 0 □□

0 = E_1 0 □

$C_2^1 \cdot C_4^2 \cdot C_8^2 = 336$

$f_{(E_3)} = \frac{2}{8} = \frac{1}{4}$

$f_{(E_2)} = \frac{2}{4} = \frac{1}{2}$

$f_{(E_1)} = \frac{1}{2}$

4 = E_3 □□0□□□□

2 = E_2 0 0 0 0

0 = E_1 □□

$C_2^0 \cdot C_4^4 \cdot C_8^1 = 8$

E_n	S_n	N_n	$C_{S_n}^{N_n}$
⋮	⋮	⋮	⋮
E_3	S_3	N_3	$C_{S_3}^{N_3}$
E_2	S_2	N_2	$C_{S_2}^{N_2}$
E_1	S_1	N_1	$C_{S_1}^{N_1}$

$N_t = \sum_i N_i$. $E_t = \sum_i N_i E_i$

$$W = \prod_i C_{S_i}^{N_i} \Rightarrow \ln W = \sum_i \ln C_{S_i}^{N_i}$$

$$\mathcal{L} = \ln W - \alpha[\Sigma N_i - N_t] - \beta[\Sigma N_i E_i - E_t]$$

$$\frac{\partial \mathcal{L}}{\partial N_i} = 0 \quad \frac{\partial \mathcal{L}}{\partial \alpha} = 0 \quad \frac{\partial \mathcal{L}}{\partial \beta} = 0$$

$$\mathcal{L} = \sum_i \ln C_{S_i}^{N_i} - \alpha[\Sigma N_i - N_t] - \beta[\Sigma N_i E_i - E_t] \qquad C_{S_i}^{N_i} = \frac{S_i!}{(S_i - N_i)!\,N_i!} \quad \ln n! \simeq n\ln n - n \text{ (stirling's approximation)}$$

$$\ln C_{S_i}^{N_i} = \ln S_i! - \ln(S_i - N_i)! - \ln(N_i!) \approx S_i \ln S_i - (S_i - N_i)\ln(S_i - N_i) - N_i \ln N_i$$

$$\frac{\partial \mathcal{L}}{\partial N_i} = [\ln(S_i - N_i) + 1 - \ln N_i - 1] - \alpha - \beta E_i = 0$$

$$= \ln\frac{S_i - N_i}{N_i} - \alpha - \beta E_i = 0$$

$$\frac{S_i - N_i}{N_i} = e^{\alpha + \beta E_i} \Rightarrow \frac{S_i}{N_i} = e^{\alpha + \beta E_i} + 1$$

$$f(E_i) = \frac{N_i}{S_i} = \frac{1}{e^{\alpha + \beta E_i} + 1} \quad f(E) = \frac{1}{e^{\alpha + \beta E} + 1}$$ Fermi function

$E \longrightarrow \infty$. $f(E) \propto e^{-\frac{E}{k_B T}}$. $k_B = 1.38 \times 10^{-23}$ J/k . T : temperature

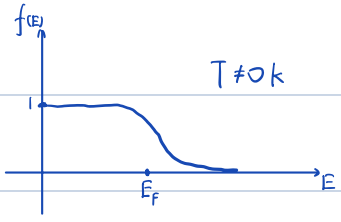
$$\Rightarrow \beta = \frac{1}{k_B \cdot T}$$

Assume when $E = E_F$, $f(E)\big|_{E=E_F} = \frac{1}{2}$

$$f(E_F) = \frac{1}{2} \quad \frac{1}{e^{\alpha + \beta E_F} + 1} = \frac{1}{2}$$

$$\Rightarrow \alpha = -\beta E_F$$

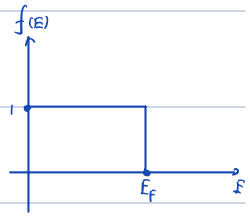
$$f(E) = \frac{1}{e^{(E - E_F)/k_B T} + 1}$$. E_F : Fermi-level 费米能级



$T = 0k$

$E < E_F$. $f(E) = 1$

$E > E_F$. $f(E) = 0$



费米能级以下被全部填满

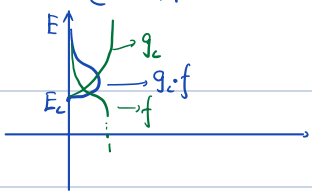
CB : $g_c(E) = \frac{m_n^* \sqrt{2m_n^*}\sqrt{E - E_c}}{\pi^2 \hbar^3}$. $E > E_c$

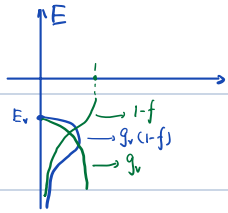
VB : $g_v(E) = \frac{m_p^* \sqrt{2m_p^*}\sqrt{E_v - E}}{\pi^2 \hbar^3}$. $E < E_v$

$f(E) = \frac{1}{e^{(E-E_f)/kT} + 1}$

$n = \int_{E_c}^{E_c^{top}} g_c(E) \cdot f(E)\, dE$

$p = \int_{E_v^{bottom}}^{E_v} g_v(E) \cdot (1 - f(E))\, dE$





CB: $n = \int_{E_c}^{E_c^{top}} g_c(E) \cdot f(E)\, dE \simeq \int_{E_c}^{\infty} \frac{m_n^* \sqrt{2m_n^*}\sqrt{E - E_c}}{\pi^2 \hbar^3} \cdot \frac{1}{1 + e^{(E-E_F)/kT}}\, dE$ assume $\eta = \frac{E - E_c}{kT}$

$= \int_0^{+\infty} \frac{m_n^* \sqrt{2m_n^*} \cdot \sqrt{kT} \cdot \sqrt{\eta}}{\pi^2 \hbar^3} \cdot \frac{1}{1 + e^{\eta} \cdot e^{\frac{E_c - E_F}{kT}}}\, kT \cdot d\eta$

$= \frac{m_n^* \sqrt{2m_n^*} \cdot (kT)^{\frac{3}{2}}}{\pi^2 \hbar^3} \cdot \int_0^{+\infty} \frac{\sqrt{\eta}}{1 + e^{\eta} \cdot e^{\frac{E_c - E_F}{kT}}}\, d\eta$.

if $E_c - E_F > 3kT$, $e^{\frac{E_c - E_F}{kT}} >> 1$

$\Rightarrow \frac{m_n^* \sqrt{2m_n^*} \cdot (kT)^{\frac{3}{2}}}{\pi^2 \hbar^3} \cdot e^{-\frac{E_c - E_F}{kT}} \cdot \int_0^{+\infty} \sqrt{\eta} \cdot e^{-\eta} \cdot d\eta$

$= \underbrace{2\left(\frac{2\pi m_n^* kT}{h^2}\right)^{\frac{3}{2}} \cdot e^{-\frac{E_c - E_F}{kT}}}_{N_c}$

$n = N_c \cdot e^{-(E_c - E_F)/kT}$

$p = \int_{E_v^{bottom}}^{E_v} g_v(E) \cdot (1 - f(E))\, dE \simeq \int_{-\infty}^{E_v} \frac{m_p^* \cdot \sqrt{2m_p^*} \cdot \sqrt{E_v - E}}{\pi^2 \hbar^3} \cdot \frac{1}{1 + e^{(E_F - E)/kT}}\, dE$

assume $\frac{E_v - E}{kT} = \eta$

$\Rightarrow \int_0^{+\infty} \frac{m_p^* \sqrt{2m_p^*} \cdot \sqrt{kT} \cdot \sqrt{\eta}}{\pi^2 \hbar^3} \cdot \frac{1}{1 + e^{\frac{E_F - E_v}{kT}} \cdot e^{\eta}} \cdot kT \cdot d\eta$. if $E_F - E_v > 3kT$, $e^{\frac{E_F - E_v}{kT}} >> 1$

$= \frac{m_p^* \sqrt{2m_p^*} \cdot (kT)^{\frac{3}{2}}}{\pi^2 \hbar^3} \int_0^{+\infty} \sqrt{\eta} \cdot e^{-\eta} \cdot d\eta \cdot e^{-\frac{E_F - E_v}{kT}}$

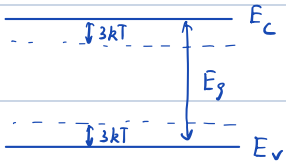
$= 2\left(\frac{2\pi m_p^* kT}{h^2}\right)^{\frac{3}{2}} \exp\left(-\frac{E_F - E_v}{kT}\right)$

$p = N_v \exp\left(-\frac{E_F - E_v}{kT}\right)$

$n = N_c \cdot \exp\left(-\frac{E_c - E_F}{kT}\right)$ $\quad E_c - E_F > 3kT$

$\Rightarrow$ non-degenerate semiconductor.

$P = N_v \cdot \exp\left(-\frac{E_F - E_v}{kT}\right)$ $\quad E_F - E_v > 3kT$



$n \times p = N_c N_v \exp\left(-\frac{E_c - E_v}{kT}\right) = N_c N_v \exp\left(-\frac{E_g}{kT}\right)$

$n = p = n_i$. intrinsic semiconductor : 本征半导体. 电子数 = 空穴数

$n_i^2 = N_c N_v \exp\left(-\frac{E_g}{kT}\right)$

$n_i = \sqrt{N_c N_v} \exp\left(-\frac{E_g}{2kT}\right)$

$N_c \exp\left(-\frac{E_c - E_{Fi}}{kT}\right) = N_v \exp\left(-\frac{E_{Fi} - E_v}{kT}\right)$

$E_{Fi} = \frac{E_c + E_v}{2} + \frac{kT}{2} \ln\frac{N_v}{N_c} \approx \frac{E_c + E_v}{2}$

本征半导体的费米能级大致在 $\frac{E_c + E_v}{2}$

$\sqrt{N_c N_v} \exp\left(-\frac{E_g}{2kT}\right) = N_c \exp\left(-\frac{E_c - E_{Fi}}{kT}\right)$

$\ln\sqrt{\frac{N_v}{N_c}} - \frac{E_g}{2kT} = -\frac{E_c - E_{Fi}}{kT}$

$E_{Fi} = E_c - \frac{E_g}{2} + kT \ln\sqrt{\frac{N_v}{N_c}}$

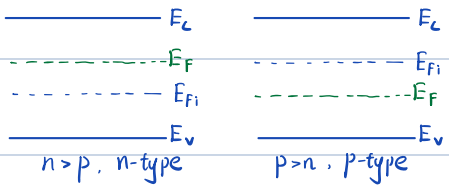
$E_{Fi} = \frac{E_c + E_v}{2} + kT \ln\sqrt{\frac{N_v}{N_c}}$

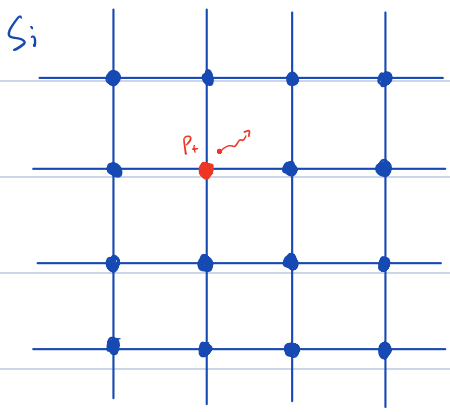
本征半导体: $n_i = N_c \cdot \exp\left(-\frac{E_c - E_{Fi}}{kT}\right)$. 非本征半导体: $n = N_c \exp\left(-\frac{E_c - E_F}{kT}\right)$

$\frac{n}{n_i} = \exp\left(\frac{E_F - E_{Fi}}{kT}\right)$.

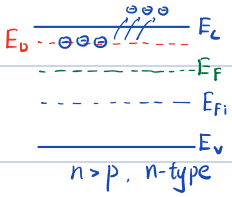
$n = n_i \exp\left(\frac{E_F - E_{Fi}}{kT}\right)$

$p \quad np = n_i^2 \quad \frac{n_i^2}{n} = n_i \exp\left(\frac{E_{Fi} - E_F}{kT}\right)$

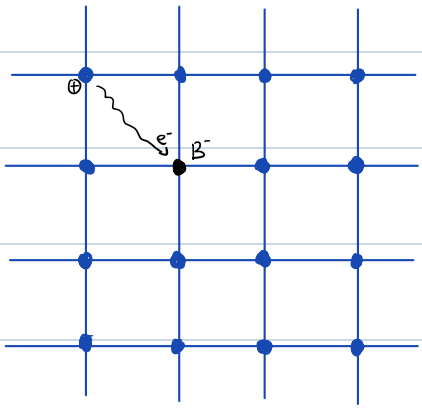




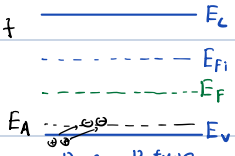
Donor : 贡献了一个电子



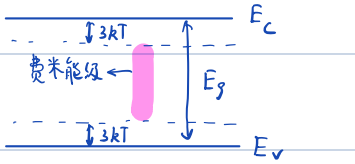
$E_1 = \frac{m^* q^4}{2(4\pi k_s \varepsilon_0 \hbar)^2} = \frac{m_0 q^4}{2(4\pi\varepsilon_0\hbar)^2} \cdot \frac{m^*}{m_0} \cdot \frac{1}{k_s^2} \approx 13.6\,\text{meV}$



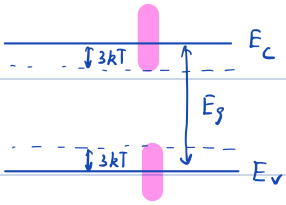
Acceptor：接收一个电子



P>n，P-type



non-degenerate semiconductor



degenerate semiconductor

(doesn't satisfies $|E_c - E_F| > 3kT$)

$$n = \int_{E_c}^{E_c+\infty} g_c(E)\cdot f(E)\,dE \simeq \int_{E_c}^{\infty} \frac{m_n^* \sqrt{2m_n^*}\sqrt{E-E_c}}{\pi^2\hbar^3}\cdot\frac{1}{1+e^{(E-E_F)/kT}}\,dE$$

$$= \int_0^{+\infty} \frac{m_n^* \sqrt{2m_n^*}\cdot\sqrt{kT}\sqrt{\eta}}{\pi^2\hbar^3}\cdot\frac{1}{1+e^{\eta}\cdot e^{\frac{E_c-E_F}{kT}}}\,kT\cdot d\eta$$

$$= \frac{m_n^* \sqrt{2m_n^*}\cdot(kT)^{\frac{3}{2}}}{\pi^2\hbar^3}\cdot\int_0^{+\infty}\frac{\sqrt{\eta}}{1+e^{\eta}\cdot e^{\frac{E_c-E_F}{kT}}}\,d\eta = \frac{m_n^* \sqrt{2m_n^*}\cdot(kT)^{\frac{3}{2}}}{\pi^2\hbar^3}\cdot\int_0^{+\infty}\frac{\sqrt{\eta}}{1+e^{\eta-\eta_F}}\,d\eta$$

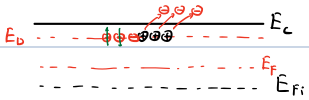
$$\eta_F = \frac{E_F - E_c}{kT}$$

$$F_{\frac{1}{2}}(\eta_F) = \int_0^{+\infty}\frac{\sqrt{\eta}}{1+e^{\eta-\eta_F}}\,d\eta$$

$$\Rightarrow n = \frac{m_n^* \sqrt{2m_n^*}\cdot(kT)^{\frac{3}{2}}}{\pi^2\hbar^3}\cdot F_{\frac{1}{2}}(\eta_F)$$

$$p = \frac{m_p^* \sqrt{2m_p^*}\cdot(kT)^{\frac{3}{2}}}{\pi^2\hbar^3}\cdot F_{\frac{1}{2}}(\eta_F'), \quad \eta_F' = \frac{E_v - E_F}{kT}$$

Donor 掺杂能级: E_D, 浓度: N_D, N_D^+, 剩余电子: $N_D - N_D^+$



每一个态只能有一个自旋方向的电子. ⇒电子之间的库伦相互作用不能同时填两个

Accept 掺杂能级: E_A, 浓度: N_A, N_A^-

$n > p$, n-type

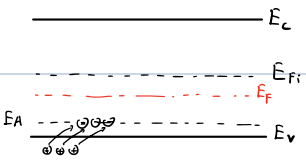
结论:

$$\frac{N_D^+}{N_D} = \frac{1}{1 + g_D e^{(E_F - E_D)/kT}} \quad (g_D = 2)$$

自旋向上, 向下

$$\frac{N_A^-}{N_A} = \frac{1}{1 + g_A e^{(E_A - E_F)/kT}} \quad (g_A = 4)$$

粒, 重空穴, 每个空穴有自旋向上, 向下.



$n < p$, p-type

推导: $W = \prod_i C_{S_i}^{N_i}$

$$\boxed{W' = W \cdot W_D} \qquad W_D = C_{N_D}^{N_D - N_D^+} \cdot g^{(N_D - N_D^+)}$$

在 E_{total}, n_{total} 不变的情况下最优化.

charge neutrality.

$$n + N_A^- = p + N_D^+$$

$$n = N_c \frac{2}{\sqrt{\pi}} F_{\frac{1}{2}}\left(\frac{E_F - E_c}{kT}\right) \qquad N_A^- = \frac{N_A}{1 + g_A e^{(E_A - E_F)/kT}}$$

$$p = N_v \frac{2}{\sqrt{\pi}} F_{\frac{1}{2}}\left(\frac{E_v - E_F}{kT}\right) \qquad N_D^+ = \frac{N_D}{1 + g_D e^{(E_F - E_D)/kT}}$$

Consider: ① complete ionization: $N_A^- = N_A$, $N_D^+ = N_D$

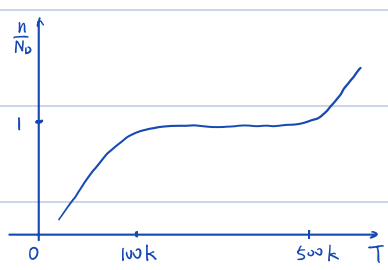
② $n \cdot p = n_i^2$

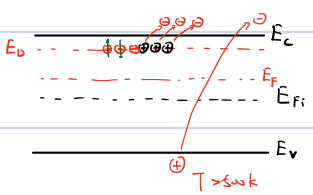
if $N_D > N_A$. $n + N_A = p + N_D$

$$\Rightarrow n = \frac{N_D - N_A}{2} + \sqrt{\left(\frac{N_D - N_A}{2}\right)^2 + n_i^2} \xrightarrow{N_D - N_A \gg n_i} N_D - N_A$$

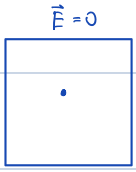
$$p \simeq \frac{n_i^2}{N_D - N_A}$$

$\frac{N_D^+}{N_D} < 1$





Carrier Transport



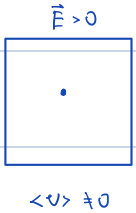
$E = \frac{3}{2}kT$. $T = 300k$, $kT = 25.9\ meV$

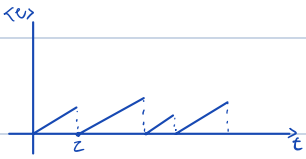
$\frac{1}{2}m^*v^2 = \frac{3}{2}kT$. $v = \sqrt{\frac{3kT}{m^*}}$ $GaAs: m^* = 0.067 m_e$

极先速度 $v \cong 4.5\times10^5\ m/s$

τ: Scattering time 散射时间.

$L = v\cdot\tau$: mean free path.





$\vec{F} = -e\cdot\vec{E}$ $\vec{a} = \frac{\vec{F}}{m_n^*} = -\frac{e}{m_n^*}\cdot\vec{E}$

$\langle\vec{v}_n\rangle = \vec{a}\cdot\tau_n = -\frac{e\tau_n}{m_n^*}\cdot\vec{E} \propto \vec{E}$

$\langle\vec{v}_n\rangle = -\mu_n\cdot\vec{E}$ μ_n: mobility for electron 迁移率

$\langle\vec{v}_p\rangle = \frac{e\tau_p}{m_p^*}\vec{E} = \mu_p\cdot\vec{E}$ μ_p: mobility for hole

$GaAs$: $\mu_n = 8500\ cm^2/V/s$

$\mu_p = 400\ cm^2/V/s$

Si : $\mu_n = 1400\ cm^2/V/s$

$\mu_p = 470\ cm^2/V/s$

$\Rightarrow \vec{J}_n = (-e)\cdot n\cdot\vec{v}_n = e\cdot n\cdot\mu_n\cdot\vec{E}$

$\vec{J}_p = (e)\cdot p\cdot\vec{v}_p = e\cdot p\cdot\mu_p\cdot\vec{E}$

$\vec{J}_{drift} = \vec{J}_n + \vec{J}_p$

$\mu = \frac{e\cdot\tau}{m^*}$.

τ: scattering time

$\frac{1}{\tau}$: scattering rate

- Lattice scattering (phonon 声子 scattering)

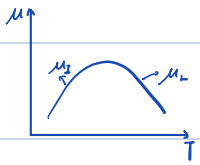
$\frac{1}{\tau_L} \propto T^{\frac{3}{2}}$ $\mu_L \propto \tau_L$ $\Rightarrow \mu_L \propto T^{-\frac{3}{2}}$

- Ioized Impurity Scattering

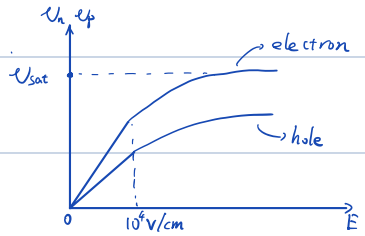
$\frac{1}{\tau_I} \propto \frac{N_I}{T^{\frac{3}{2}}}$. $N_I = N_D^+ + N_A^-$ $\mu_I \propto \frac{T^{\frac{3}{2}}}{N_I}$

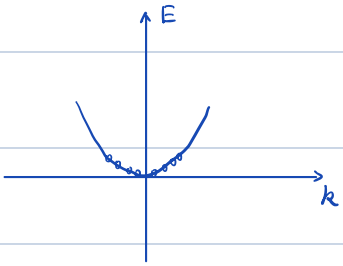
$$\frac{1}{\tau} = \frac{1}{\tau_L} + \frac{1}{\tau_I} + \cdots$$

$$\mu \propto \tau \rightarrow \frac{1}{\mu} = \frac{1}{\mu_L} + \frac{1}{\mu_I} + \cdots$$



$v_n = \mu_n \cdot \vec{E}$

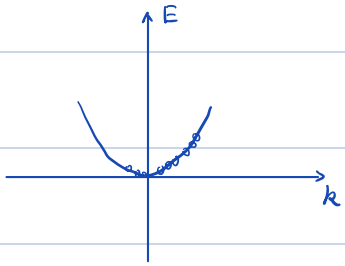




$|\vec{E}| = 0$

$v = 0$

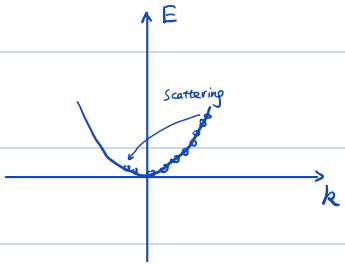
$J = 0$



$|\vec{E}| \ll \mathcal{E}_{sat}$ (比较电场)

$v > 0$

$J \neq 0$

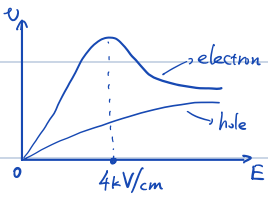


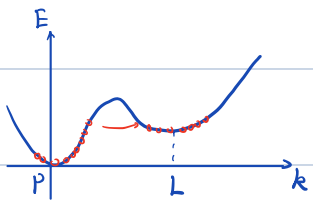
$|\vec{E}| \cong \mathcal{E}_{sat}$

$|\vec{E}| > \mathcal{E}_{sat}$

$v \longrightarrow v_{sat}$

GaAs

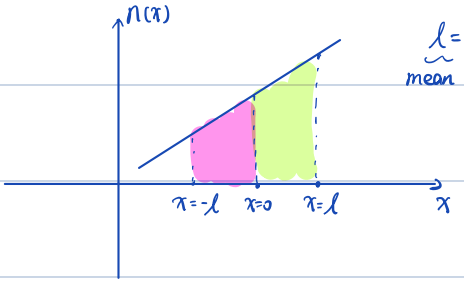




$m^*_{n,L} > m^*_{n,\Gamma}$

$E > \mathcal{E}_{sat}$, $m^* \nearrow \rightarrow v \downarrow$

diffusion current



$\underbrace{l = v \cdot \tau}$ mean free path

两个区域的电子: $\begin{cases} \frac{n(-l)+n(0)}{2} \cdot l \cdot A \\ \frac{n(l)+n(0)}{2} \cdot l \cdot A \end{cases}$

在不施加外部电场时，一半向左运动，一半向右运动.

$$F_n = \frac{A\left[\frac{n(-l)+n(0)}{2} \cdot l \cdot \frac{1}{2} - \frac{n(l)+n(0)}{2} \cdot l \cdot \frac{1}{2}\right]}{A \cdot \frac{l}{v}}$$

$$F_n = \frac{n(-l)-n(l)}{4} \cdot v \xrightarrow{l \text{ is small}} \frac{1}{4} \cdot \frac{\partial n}{\partial x}\Big|_{x=0} \cdot (-2l) \cdot v$$

$$F_n = -\frac{l \cdot v}{2} \cdot \frac{\partial n}{\partial x}$$

$$\Rightarrow \begin{cases} F_n = -D_n \cdot \frac{\partial n}{\partial x} & D_n = \frac{l_n \cdot v_n}{2} \\ F_p = -D_p \cdot \frac{\partial p}{\partial x} & D_p = \frac{l_p \cdot v_p}{2} \end{cases}$$ → diffusion coefficient

$$J_n^{diff} = (-e) \cdot F_n = e \cdot D_n \cdot \frac{\partial n}{\partial x} \qquad \vec{J}_n^{diff} = e \cdot D_n \cdot \nabla n$$

$$J_p^{diff} = (+e) \cdot F_n = -e D_p \cdot \frac{\partial p}{\partial x} \qquad \vec{J}_p^{diff} = -e \cdot D_p \cdot \nabla p$$

$$\begin{cases} \vec{J}_n = \vec{J}_n^{diff} + \vec{J}_n^{drift} = e n \cdot \mu_n \cdot E + e \cdot D_n \cdot \frac{\partial n}{\partial x} \\ \vec{J}_p = \vec{J}_p^{diff} + \vec{J}_p^{drift} = e p \cdot \mu_p \cdot E + (-e) \cdot D_p \cdot \frac{\partial p}{\partial x} \\ \vec{J} = \vec{J}_n + \vec{J}_p \end{cases}$$

$$\vec{J}_n^{drift} = e n \cdot \mu_n \cdot \vec{E}$$

$$\vec{J}_p^{drift} = e p \cdot \mu_p \cdot \vec{E}$$

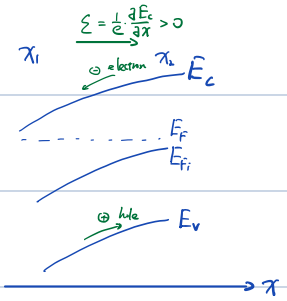
1D: $D_n = \frac{l_n \cdot v_n}{2}$ $\quad D_p = \frac{l_p \cdot v_p}{2}$

3D: $D_n = \frac{l_n \cdot v_n}{3}$ $\quad D_p = \frac{l_p \cdot v_n}{3}$

$$\frac{D_n}{\mu_n} = \frac{\frac{l_n \cdot v_n}{3}}{\frac{e \tau_n}{m_n^*}} = \frac{\frac{v_n^2 \cdot \tau_n}{3}}{\frac{e \cdot \tau_n}{m_n^*}} = \frac{\frac{1}{3} m_n^* v_n^2}{e} \xrightarrow{\frac{1}{2} m_n^* v_n^2 = \frac{3}{2} kT} \frac{kT}{e} \approx 25.9\text{mV}$$ Einstein Relationship

$$\frac{D_p}{\mu_p} = \frac{kT}{e}$$

n-type. $n(x_1) > n(x_2)$. $N_d(x_1) > N_d(x_2)$



$$E_c(x) = E_{ref} + (-e)\cdot V(x) \quad . \quad E_v(x) = E_{ref} + (-e)\cdot V(x) - E_g$$

electric field $\mathcal{E}$

$$\mathcal{E} = -\frac{\partial V}{\partial x} = -\frac{\partial\left(\frac{E_c}{-e}\right)}{\partial x}$$

$$\boxed{\mathcal{E} = \frac{1}{e}\cdot\frac{\partial E_c}{\partial x} = \frac{1}{e}\cdot\frac{\partial E_v}{\partial x} = \frac{1}{e}\cdot\frac{\partial E_{Fi}}{\partial x}}$$

$$J_n = J_n^{drift} + J_n^{diff} \equiv 0$$

$$J_n^{drift} = e\cdot n\cdot\mu_n\cdot\mathcal{E} \qquad J_n^{diff} = e\cdot D_n\cdot\frac{\partial n}{\partial x}$$

$$n = n_i\cdot\exp\left(\frac{E_F - E_{Fi}(x)}{kT}\right) \quad . \quad \frac{\partial n}{\partial x} = n_i\cdot\exp\left(\frac{E_F - E_{Fi}}{kT}\right)\left(-\frac{\partial E_{Fi}}{\partial x}\right)\cdot\left(\frac{1}{kT}\right)$$

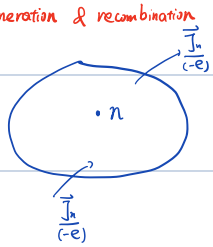
$$\frac{\partial n}{\partial x} = n\cdot(-e\cdot\mathcal{E})\cdot\frac{1}{kT}$$

$$e\cdot n\cdot\mu_n\cdot\mathcal{E} + e\cdot D_n\cdot n\cdot(-e\cdot\mathcal{E})\cdot\frac{1}{kT} = 0$$

$$\frac{D_n}{\mu_n} = \frac{kT}{e}$$

$$\begin{cases}\vec{J}_n = e\cdot n\cdot\mu_n\vec{E} + e\cdot D_n\cdot\nabla n\\ \vec{J}_p = e\cdot p\cdot\mu_p\vec{E} - e\cdot D_p\cdot\nabla p\end{cases} \qquad \vec{J} = \vec{J}_n + \vec{J}_p$$

no generation & recombination



$$N = \int_V n\,dV$$

$$\frac{d}{dt}\int_V n\cdot dV = \frac{dN}{dt} = -\oint\frac{\vec{J}_n}{(-e)}\cdot d\vec{S} = \frac{1}{e}\cdot\iiint\nabla\cdot\vec{J}_n\,dV$$

$$\frac{\partial n}{\partial t} = \frac{1}{e}\cdot\nabla\cdot\vec{J}_n$$

$$\frac{\partial p}{\partial t} = -\frac{1}{e}\cdot\nabla\cdot\vec{J}_p$$

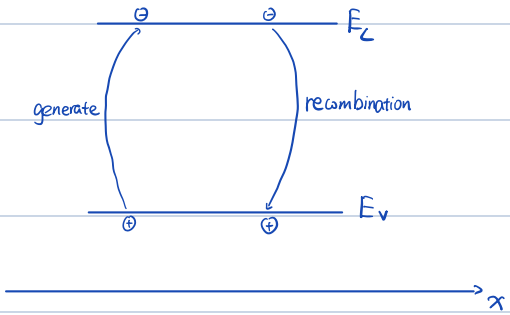
$$\begin{cases}\vec{J}_n = e\cdot n\cdot\mu_n\vec{E} + e\cdot D_n\cdot\nabla n\\ \vec{J}_p = e\cdot p\cdot\mu_p\vec{E} - e\cdot D_p\cdot\nabla p\\ \frac{\partial n}{\partial t} = \frac{1}{e}\cdot\nabla\cdot\vec{J}_n\\ \frac{\partial p}{\partial t} = -\frac{1}{e}\cdot\nabla\cdot\vec{J}_p\\ \nabla\cdot\underbrace{\vec{D}}_{\vec{D}=\varepsilon_r\varepsilon\cdot\vec{E}} = \rho = e\cdot(p + N_D^+ - n - N_A^-)\end{cases}$$

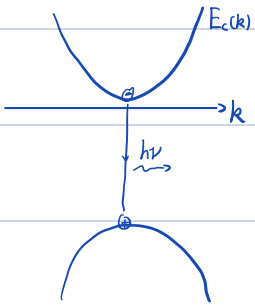
Generation and Recombination

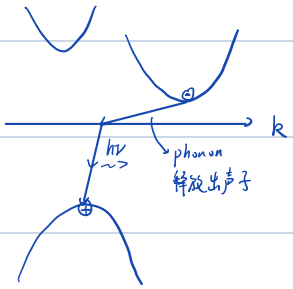
产生率: g. → $1/cm^3/s$

复合率: r → $1/cm^3/s$

① Band to Band generation & recombination







clirect

semiconductor

g. $g = A$

$r \propto n$, $r \propto p$ $\Rightarrow r = B \cdot n \cdot p$

平衡状态下

$n = n_0$, $p = p_0$. $n_0 p_0 = n_i^2$

$\left.\frac{dn}{dt}\right|_{n=n_0} = g - r = 0 \Rightarrow A = B \cdot n_0 p_0 = B \cdot n_i^2$

$\frac{dn}{dt} = g - r = Bn_i^2 - Bnp$

$\Rightarrow g_{net} = \frac{dn}{dt} = B(n_i^2 - np)$

$np > n_i^2$. $\frac{dn}{dt} < 0$ recombination

$np < n_i^2$. $\frac{dn}{dt} > 0$ generation

Dark:

$n: n_0 \longrightarrow n = n_0 + \delta n$ $\delta n = \delta p$. assume p-type. $p_0 >> n_0$

$p: p_0 \longrightarrow p = p_0 + \delta p$

$g_{net} = \frac{dn}{dt} = B[n_i^2 - (n_0 p_0 + \delta n \cdot (n_0 + p_0) + \delta n \cdot \delta p)]$

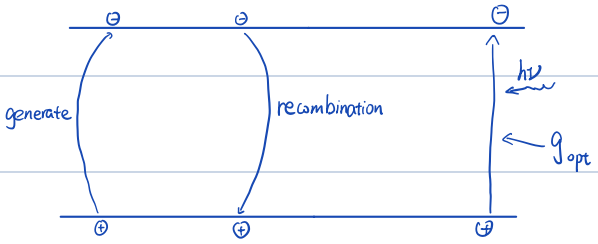
$p_0 \gg \delta n, \delta p$ (low level injection) $\Rightarrow g_{net} \cong -B \cdot \delta n \cdot (n_0 + p_0) \overset{p_0 >> n_0}{\approx} -B \cdot p_0 \cdot \delta n$

$\frac{d\delta n}{dt} = -B p_0 \cdot \delta n = -\frac{\delta n}{\tau}$

$\tau = \frac{1}{B p_0}$ $\delta n = \delta n(0) \cdot e^{-\frac{t}{\tau}}$

carrier lifetime

light is on :



$$g_{net} = \frac{\partial n}{\partial t} = g - r + g_{opt}$$

$$= -BP_0 \cdot \delta n + g_{opt}$$

$$\frac{d\delta n}{dt} = -\frac{\delta n}{\tau} + g_{opt} \quad \tau = \frac{1}{BP_0}$$

Steady state : $\frac{d\delta n}{dt} = 0$. $-\frac{\delta n}{\tau} + g_{opt} = 0$

$$\delta n = g_{opt} \cdot \tau$$

assume Si . $P_0 = N_A = 10^{16}\,cm^{-3}$. $n_i = 10^{10}\,cm^{-3}$. $g_{opt} = 10^{20}/cm^3/s$. $\tau = 1\mu s$.

$n_0 = \frac{n_i^2}{P_0} = 10^4\,cm^{-3}$

$\delta n = g_{opt} \cdot \tau = 10^{14}\,cm^{-3}$

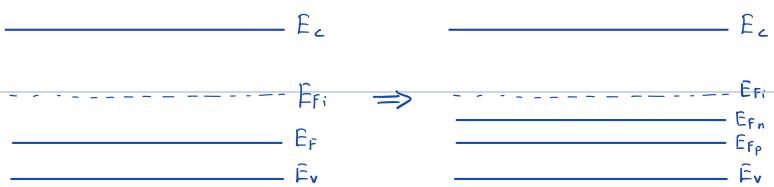
$P = P_0 + \delta n = (10^{16} + 10^{14})\,cm^{-3} \approx 10^{16}\,cm^{-3}$

$n = n_0 + \delta n = (10^4 + 10^{14})\,cm^{-3} \approx 10^{14}\,cm^{-3}$

$$\Rightarrow p \cdot n \neq n_i^2$$

$n = n_i \exp\left(\frac{E_{Fn} - E_{Fi}}{kT}\right)$ E_{Fn} . E_{Fp} : Quasi-fermi-level 准费米能级

$P = n_i \cdot \exp\left(\frac{E_{Fi} - E_{Fp}}{kT}\right)$ $n \cdot p = n_i^2 \cdot \exp\left(\frac{E_{Fn} - E_{Fp}}{kT}\right)$

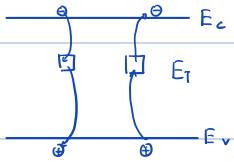


$$\frac{\partial n}{\partial x} = n \cdot \frac{1}{kT}\left[\frac{\partial E_{Fn}}{\partial x} - \frac{\partial E_{Fi}}{\partial x}\right] = n \cdot \frac{1}{kT}\left[\frac{\partial E_{Fn}}{\partial x} - eE\right] \quad . \quad \frac{D_n}{\mu_n} = \frac{kT}{e}$$

$$J_n = J_n^{drift} + J_n^{diff} = en\mu_n \vec{E} + eD_n \frac{\partial n}{\partial x} = en\mu_n E + kT\mu_n \left[n \cdot \frac{1}{kT}\left(\frac{\partial E_{Fn}}{\partial x} - eE\right)\right]$$

$$\Rightarrow \begin{cases} J_n = n\mu_n \frac{\partial E_{Fn}}{\partial x} \\ J_p = p\mu_p \frac{\partial E_{Fp}}{\partial x} \end{cases}$$

① Shockley - Read - Hall (SRH) G&R



steady state. net generation rate. g_{net}

net recombination rate. $r_{net} = -g_{net}$

$$r_{net} = \frac{(pn - n_i^2)}{Z_p \cdot (n + n_i \exp(\frac{E_T - E_i}{kT})) + Z_n (p + n_i \exp(\frac{E_i - E_T}{kT}))}$$

$pn > n_i^2$. $r_{net} > 0$

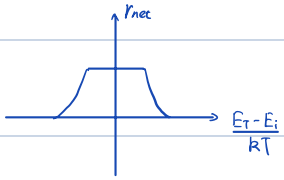
$pn < n_i^2$. $r_{net} < 0$. $g_{net} > 0$

E_i: 本征费米能级

E_T: 缺陷能级

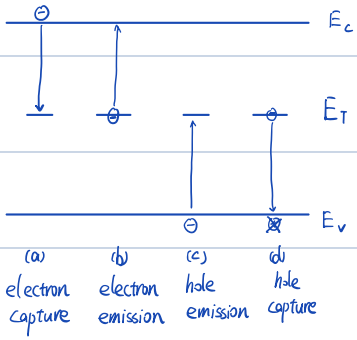
assume: $Z_p = Z_n$

when $E_T = E_i = E_{Fi}$, r_{net} maximized.



Traps near mid-gap are much more efficient as recombination center

推导 r_{net}.



假设缺陷浓度 $N_T = n_T + p_T$

n_T: electron on trap

p_T: no electron on trap

$$\frac{\partial n}{\partial t} = \left(\frac{\partial n}{\partial t}\right)_b - \left(\frac{\partial n}{\partial t}\right)_a \qquad \left(\frac{\partial n}{\partial t}\right)_a = C_n \cdot n \cdot p_T$$

$$\frac{\partial p}{\partial t} = \left(\frac{\partial p}{\partial t}\right)_c - \left(\frac{\partial p}{\partial t}\right)_d \qquad \left(\frac{\partial n}{\partial t}\right)_b = e_n \cdot n_T$$

$$\frac{\partial n}{\partial t} = e_n n_T - C_n \cdot n \cdot p_T$$

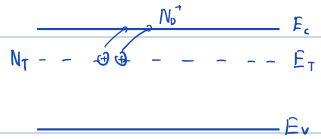
$$\frac{\partial p}{\partial t} = e_p p_T - C_p \cdot p \cdot n_T$$

$$\Rightarrow \begin{cases} e_n n_{T_0} = C_n \cdot n_0 \cdot p_{T_0} & e_n = C_n \underbrace{n_0 \frac{p_{T_0}}{n_{T_0}}}_{n_1} \\ e_p \cdot p_{T_0} = C_p \cdot p_0 \cdot n_{T_0} & e_p = C_p \underbrace{p_0 \frac{n_{T_0}}{p_{T_0}}}_{p_1} \end{cases}$$

when $n = n_0$, $p = p_0$, $n_T = n_{T_0}$, $p_T = p_{T_0}$, $\frac{\partial n}{\partial t} = \frac{\partial p}{\partial t} = 0$

$$\frac{\partial n}{\partial t} = C_n n_1 - C_n \cdot n \cdot p_T$$

$$\frac{\partial p}{\partial t} = C_p \cdot p_1 - C_p \cdot p \cdot n_T$$



if N_T : donor like.

$N_D^+ \longrightarrow P_{T_0}$

$N_D - N_D^+ \longrightarrow n_{T_0}$

$$\frac{n_{T_0}}{P_{T_0}} = g_T \cdot \exp\left(\frac{E_F - E_T}{kT}\right) = \exp\left(\frac{E_F - E_T'}{kT}\right)$$

$$\begin{cases} P_1 = P_0 \cdot \frac{n_{T_0}}{P_{T_0}} = n_i \cdot \exp\left(\frac{E_{Fi} - E_F}{kT}\right) \cdot \exp\left(\frac{E_F - E_T'}{kT}\right) = n_i \cdot \exp\left(\frac{E_{Fi} - E_T'}{kT}\right) \\ n_1 = n_0 \cdot \frac{P_{T_0}}{n_{T_0}} = n_i \exp\left(\frac{E_T' - E_{Fi}}{kT}\right) \end{cases}$$

$$\frac{\partial n}{\partial t} = C_n n_1 - C_n \cdot n \cdot P_T \qquad P_1 = n_i \cdot \exp\left(\frac{E_{Fi} - E_T'}{kT}\right)$$

$$\frac{\partial P}{\partial t} = C_p \cdot P_1 - C_p \cdot P \cdot n_T \qquad n_1 = n_i \exp\left(\frac{E_T' - E_{Fi}}{kT}\right)$$

Steady state : $\frac{\partial n_T}{\partial t} = 0 \qquad \frac{\partial n}{\partial t} = \frac{\partial P}{\partial t}$

$$\begin{cases} C_n n_1 - C_n \cdot n \cdot P_T = C_p \cdot P_1 - C_p \cdot P \cdot n_T \\ N_T = n_T + P_T \end{cases}$$

$$\Rightarrow n_T = \frac{C_n \cdot n + C_p \cdot P_1}{C_n (n + n_1) + C_p (P + P_1)} \cdot N_T \qquad \text{有电子的缺陷数目}$$

$$r_{net} = \frac{\partial n}{\partial t} = \frac{nP - n_i^2}{\frac{1}{C_p \cdot N_T} \cdot (n + n_1) + \frac{1}{C_n \cdot N_T} \cdot (P + P_1)} \qquad \frac{1}{C_p \cdot N_T} = Z_p \qquad \frac{1}{C_n \cdot N_T} = Z_n \qquad P_1 = n_i \cdot \exp\left(\frac{E_{Fi} - E_T'}{kT}\right) \qquad n_1 = n_i \exp\left(\frac{E_T' - E_{Fi}}{kT}\right)$$

a) low - level injection

$n = n_0 + \delta n$

$P = P_0 + \delta P$ $\qquad P$- type. $\qquad \delta n, \delta P \ll P_0. \qquad n_0 \ll P_0$

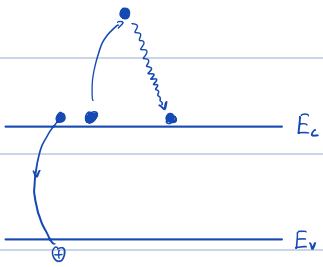
$$r_{net} = \frac{(n_0 + \delta n)(P_0 + \delta P) - n_i^2}{Z_p (n_0 + \delta n + n_1) + Z_n (P_0 + \delta P + P_1)} = \frac{P_0 \cdot \delta n}{Z_p \cdot n_1 + Z_n (P_0 + P_1)} \xrightarrow[n_1 = n_i,\ P_1 = P_i]{\text{assume } E_T = E_{Fi}} \frac{\delta n}{Z_n} \quad \cdots \quad P\text{-type}$$

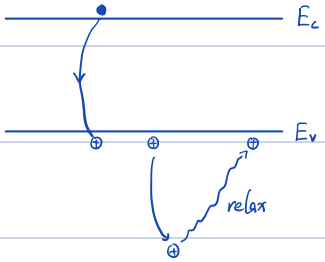
$$r_{net} = \frac{\delta P}{Z_p} \quad \cdots \quad n\text{-type}$$

b). $n \sim 0, \quad P \sim 0$

$$r_{net} = -\frac{n_i^2}{Z_p \cdot n_1 + Z_n \cdot P_1} \xrightarrow[n_1 = n_i,\ P_1 = n_i]{E_T = E_{Fi}} -\frac{n_i}{Z_n + Z_p} = -g_{net}$$

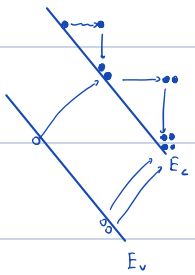
③ Auger Recombination





$$\frac{1}{\tau_{Auger}} = B \cdot n_0^2 = B \cdot N_D^2$$

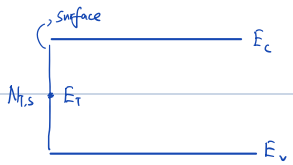
Impact ionization



电子加速，碰撞阻挡，产生新的电子/空穴对

$e \longrightarrow 2e + h$

④ surface generation & recombination



$$r_N = C_n \cdot P_T \cdot n - e_n \cdot n_T$$

$$r_P = C_p \cdot n_T \cdot P - e_p \cdot P_T$$

$$r_{net} = \frac{\partial n}{\partial t} = \frac{nP - n_i^2}{\frac{1}{C_p \cdot N_T} \cdot (n + n_1) + \frac{1}{C_n \cdot N_T} \cdot (P + P_1)}$$

$N_{T,s} \rightsquigarrow 1/cm^2$. $n, P, n_1, P_1 \rightsquigarrow 1/cm^3$

$n_T, P_T, N_T \rightsquigarrow 1/cm^3$. $e_n, e_p \rightsquigarrow 1/s$. $C_n, C_p \rightsquigarrow cm^3/s$

$r_N, r_P \rightsquigarrow 1/cm^3/s$

$$r_{N,S} = C_{ns} \cdot P_{Ts} \cdot n_s - e_{ns} \cdot n_{Ts}$$

$$r_{PS} = C_{ps} \cdot n_{Ts} \cdot P_s - e_{ps} \cdot P_{Ts}$$

$$r_{s,net} = \frac{\partial n}{\partial t} = \frac{n_s P - n_i^2}{\frac{1}{C_{ps} \cdot N_{Ts}} \cdot (n_s + n_{1s}) + \frac{1}{C_{ns} N_{Ts}} \cdot (P_s + P_{1s})}$$

$n_s, P_s, n_{1s}, P_{1s} \rightsquigarrow 1/cm^3$

$n_T, P_T, N_T \rightsquigarrow 1/cm^2$

$e_{n,s}, e_{p,s} \rightsquigarrow 1/s$

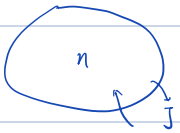
$C_{n,s}, C_{p,s} \rightsquigarrow cm^2/s$

$\frac{1}{C_{ps} N_{Ts}} \rightsquigarrow s/cm$

$r_{Ns}, r_{Ps} \rightsquigarrow 1/cm^2/s$

With G & R

$$\begin{cases} \vec{J}_n = e n \mu_n \vec{E} + e D_n \nabla n \\ \vec{J}_p = e p \mu_p \vec{E} - e D_p \nabla p \\ \frac{\partial n}{\partial t} = \frac{1}{e} \nabla \cdot \vec{J}_n + g_n - r_n \\ \frac{\partial p}{\partial t} = -\frac{1}{e} \nabla \cdot \vec{J}_p + g_p - r_p \\ \nabla \cdot \vec{D} = \rho = e(p + N_D^+ - n - N_A^-) \\ \vec{D} = \varepsilon_r \varepsilon_0 \vec{E} \end{cases}$$



$$\frac{d}{dt}\int n \, dv = \underbrace{-\oint_{(cs)} \vec{J} \cdot d\vec{s}}_{\frac{1}{e}\int_v \nabla \cdot \vec{J} \, dv} + \int g \, dv - \int r \, dv$$

$$\Rightarrow \frac{\partial n}{\partial t} = \frac{1}{e} \nabla \cdot \vec{J}_n + g_n - r_n$$

1D.

$$\begin{cases} J_n = e n \mu_n E + e D_n \frac{\partial n}{\partial x} \\ J_p = e p \mu_p E - e D_p \frac{\partial p}{\partial x} \\ \frac{\partial n}{\partial t} = \frac{1}{e} \frac{\partial}{\partial x} J_n + g_n - r_n \\ \frac{\partial p}{\partial t} = -\frac{1}{e} \frac{\partial}{\partial x} J_p + g_p - r_p \\ \frac{\partial}{\partial x} D = \rho, \quad D = \varepsilon_r \varepsilon_0 E \end{cases}$$

P-type semiconductor

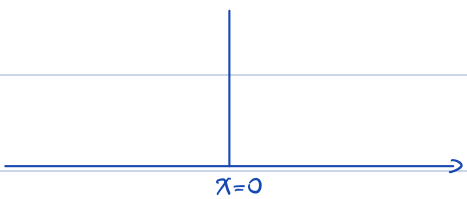
$p_0 \gg n_0$. $E \approx 0$. Quasi-Neutral Region

$e n \mu_n E$ ⟶ small.

$J_n = e D_n \frac{\partial n}{\partial x}$

$\frac{\partial n}{\partial t} = D_n \frac{\partial n^2}{\partial x^2} + g_n - r_n$, $n = n_0 + \delta n$. $g_{net} = g_n - r_n = -\frac{\delta n}{\tau_n}$

$$\Rightarrow \frac{\partial \delta n}{\partial t} = D_n \frac{\partial^2 \delta n}{\partial x^2} - \frac{\delta n}{\tau_n}$$



$\delta n(x=0)$

Steady state: $\frac{\partial \delta n}{\partial t} = 0$. $D_n \frac{\partial^2 \delta n}{\partial x^2} - \frac{\delta n}{\tau_n} = 0$

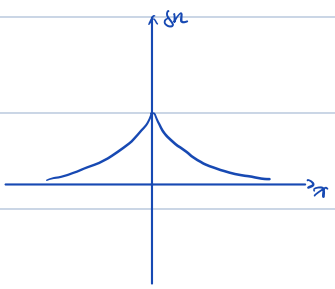
$$\delta n(x) = A e^{\frac{x}{L_n}} + B e^{-\frac{x}{L_n}}, \quad L_n = \sqrt{D_n \tau_n}$$

$\delta n(0)$ is known, $x \longrightarrow \infty$. $\delta n(\infty) \longrightarrow 0$

$x \longrightarrow -\infty$, $\delta n(-\infty) \longrightarrow 0$

$x > 0$. $\delta n(x) = \delta n(0) \cdot e^{-\frac{x}{L_n}}$

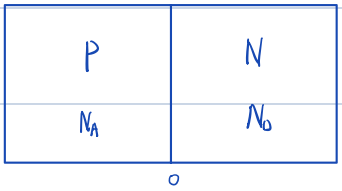
$x < 0$, $\delta n(x) = \delta n(0) \cdot e^{\frac{x}{L_n}}$

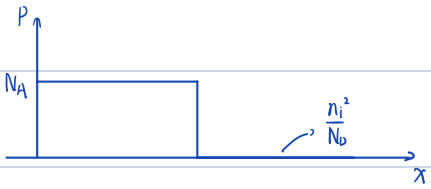


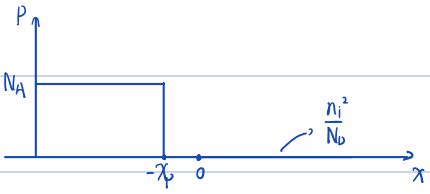
PN Junction

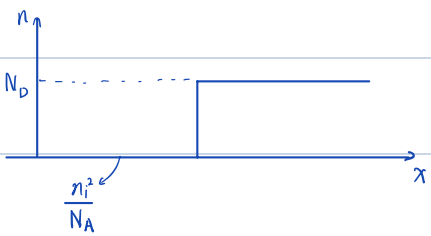
扩散前

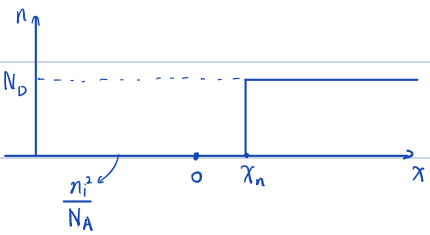
扩散

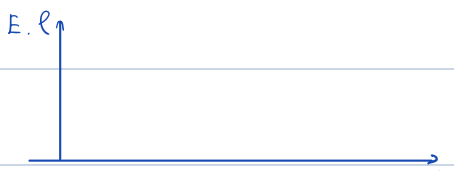






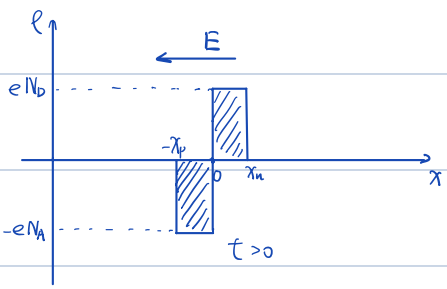






$E \approx 0, \quad \rho \approx 0$

$t < 0$



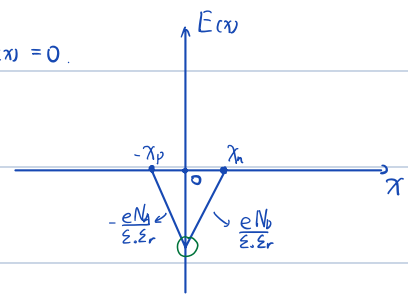
$t > 0$

depletion region Approximation

$$\rho(x) = \begin{cases} -eN_A, & -x_p \le x < 0 \\ eN_D, & 0 < x \le x_n \end{cases}$$

Poisson expression: $\frac{\partial E}{\partial x} = \frac{\rho}{\varepsilon_0 \varepsilon_r}$, $\frac{\partial D}{\partial x} = \rho$

for $x < -x_p$, $x > x_n$, $E(x) = 0$.

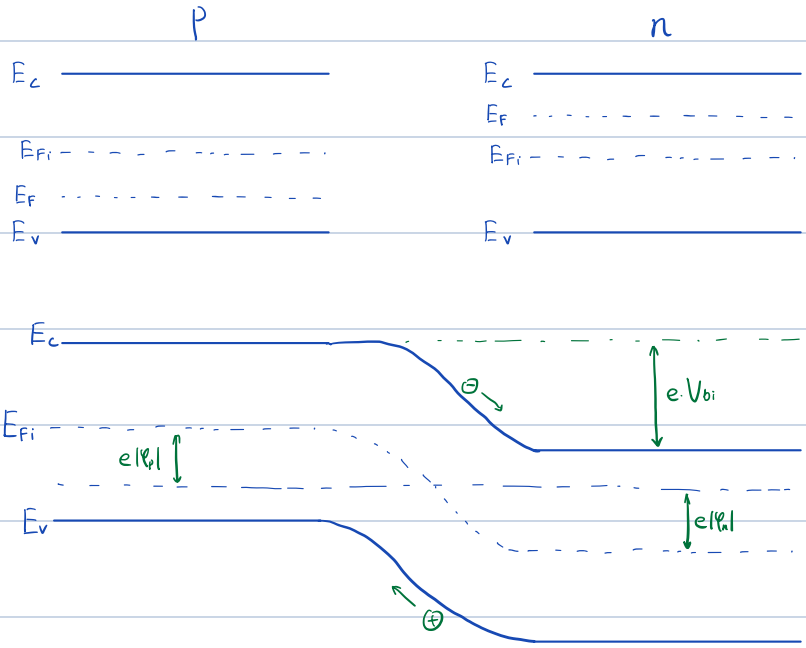


$$E(x) = \begin{cases} 0, & x \le -x_p, \ x \ge x_n \\ -\frac{eN_A}{\varepsilon_0\varepsilon_r}(x + x_p), & -x_p \le x < 0 \\ \frac{eN_D}{\varepsilon_0\varepsilon_r}(x - x_n), & 0 < x \le x_n \end{cases}$$

$$\int_{0^-}^{0^+} \frac{\partial D}{\partial x}\,dx = \int_{0^-}^{0^+} \rho\,dx = 0 \quad \Rightarrow D(0^+) - D(0^-) = 0 \quad \Rightarrow \quad -eN_A x_p = -eN_D x_n$$

若有面电荷密度，则 $D(0^+) - D(0^-) \neq 0$

$$\Rightarrow \boxed{N_A \cdot x_p = N_D \cdot x_n}$$



$$n = N_D = n_i \exp\left(\frac{E_F - E_{Fi}}{kT}\right), \quad n = n_i \cdot \exp\left(\frac{e|\varphi_n|}{kT}\right)$$

$$|\varphi_n| = \frac{kT}{e} \cdot \ln\left(\frac{N_D}{n_i}\right)$$

$$P = N_A = n_i \exp\left(\frac{E_{Fi} - E_F}{kT}\right), \quad P = n_i \exp\left(\frac{e|\varphi_p|}{kT}\right)$$

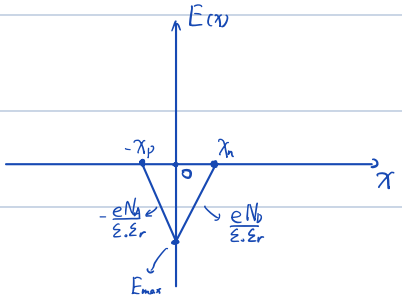
$$|\varphi_p| = \frac{kT}{e} \cdot \ln\left(\frac{N_A}{n_i}\right)$$

$$V_{bi} = |\varphi_p| + |\varphi_n| = \frac{kT}{e} \cdot \ln\left(\frac{N_A \cdot N_D}{n_i^2}\right)$$

$$V_{bi} = \frac{E_c(-x_p) - E_c(x_n)}{e}$$

$$V_{bi} = -\int_{-x_p}^{x_n} E \, dx = \frac{E_{max} \cdot x_p}{2} + \frac{E_{max} \cdot x_n}{2}$$

$$E_{max} = \frac{eN_A}{\varepsilon_0 \varepsilon_r} x_p = \frac{eN_D}{\varepsilon_0 \varepsilon_r} x_n$$

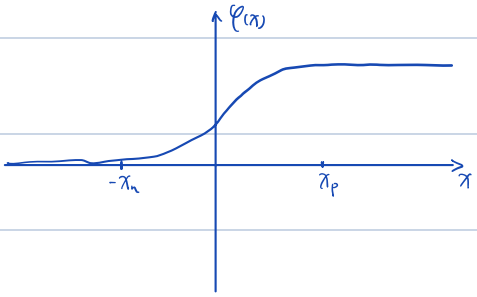


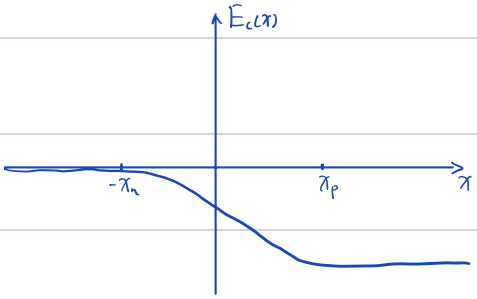
$$x_p = \frac{E_{max} \cdot \varepsilon_0 \varepsilon_r}{eN_A} \qquad x_n = \frac{E_{max} \cdot \varepsilon_0 \varepsilon_r}{eN_D}$$

$$\frac{1}{2} \cdot E_{max}^2 \left[\frac{\varepsilon_0 \varepsilon_r}{eN_A} + \frac{\varepsilon_0 \varepsilon_r}{eN_D}\right] = V_{bi}, \quad E_{max} = \sqrt{\frac{2eV_{bi}}{\varepsilon_0 \varepsilon_r}\left(\frac{N_A N_D}{N_A + N_D}\right)}$$

$$\begin{cases} x_p = \dfrac{E_{max} \cdot \varepsilon_0 \varepsilon_r}{eN_A} \\ x_n = \dfrac{E_{max} \cdot \varepsilon_0 \varepsilon_r}{eN_D} \\ E_{max} = \sqrt{\dfrac{2eV_{bi}}{\varepsilon_0 \varepsilon_r}\left(\dfrac{N_A N_D}{N_A + N_D}\right)} \end{cases} \Rightarrow \begin{cases} x_n = \sqrt{\dfrac{2\varepsilon_0 \varepsilon_r V_{bi}}{e}\left(\dfrac{N_A}{N_D(N_A+N_D)}\right)} \\ x_p = \sqrt{\dfrac{2\varepsilon_0 \varepsilon_r V_{bi}}{e}\left(\dfrac{N_D}{N_A(N_A+N_D)}\right)} \end{cases}, \quad w = x_n + x_p$$

$$\varphi(x) = -\int_{-\infty}^{x} E(x') \, dx' = -\int_{-x_p}^{x} E(x') \, dx = \begin{cases} \dfrac{eN_A}{2\varepsilon_0 \varepsilon_r}(x + x_p)^2, & -x_p < x < 0 \\ -\dfrac{eN_D}{2\varepsilon_0 \varepsilon_r}(x - x_n)^2 + V_{bi}, & 0 < x \leq x_n \\ V_{bi} = \dfrac{eN_A x_p^2 + eN_D x_n^2}{2\varepsilon_0 \varepsilon_r}, & x > x_n \end{cases}$$

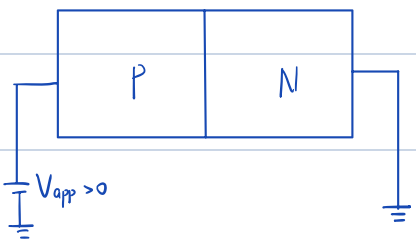




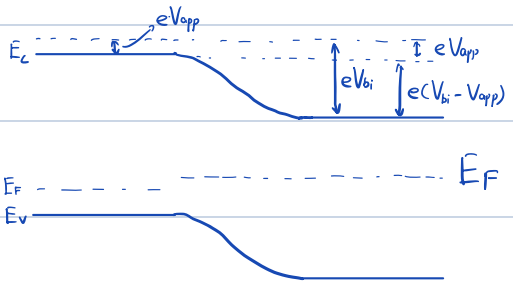
$$E_c(x) = E_{ref} + (-e)\cdot\varphi(x)$$

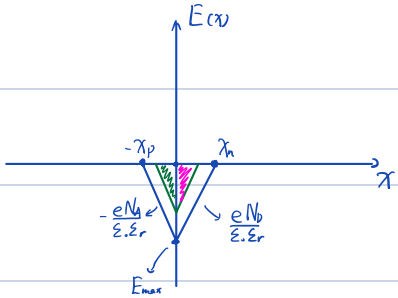
$$E_{ref} = E_c(-\infty)$$

$$E_v(x) = E_c(x) - E_g$$



$$E_c(x) = E_{ref} + (-e)\cdot\varphi(x)$$

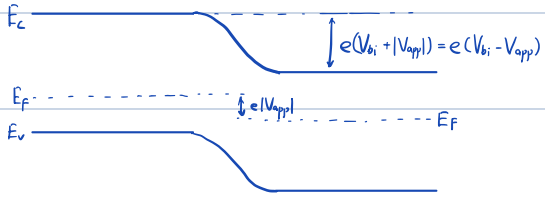


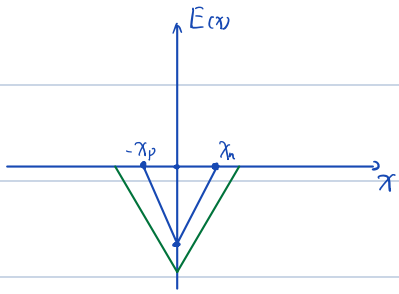


$$S_\Delta = \frac{1}{2}\left(E'_{max}\cdot x_p' + E_{max}\cdot x_n'\right) = V_{bi} - V_{app}$$

$$\Rightarrow \begin{cases} E'_{max} = \sqrt{\dfrac{2e(V_{bi}-V_{app})}{\varepsilon_0\varepsilon_r}\left(\dfrac{N_A\cdot N_D}{N_A+N_D}\right)} \\ x_n = \sqrt{\dfrac{2\varepsilon_0\varepsilon_r(V_{bi}-V_{app})}{e}\left(\dfrac{N_D}{N_A(N_A+N_D)}\right)} \\ x_p = \sqrt{\dfrac{2\varepsilon_0\varepsilon_r(V_{bi}-V_{app})}{e}\left(\dfrac{N_A}{N_D(N_A+N_D)}\right)} \end{cases}$$

$$E_c(x) = E_{ref} + (-e)\cdot\varphi(x)$$

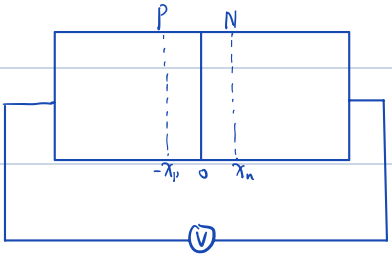




$$S_\Delta = \frac{1}{2}(E'_{max}\cdot x_p' + E_{max}\cdot x_n') = V_{bi} - V_{app}$$

$$\Rightarrow \begin{cases} E'_{max} = \sqrt{\dfrac{2e(V_{bi} - V_{app})}{\varepsilon_0 \varepsilon_r}\left(\dfrac{N_A \cdot N_D}{N_A + N_D}\right)} \\ x_n = \sqrt{\dfrac{2\varepsilon_0\varepsilon_r(V_{bi} - V_{app})}{e}\left(\dfrac{N_D}{N_A(N_A+N_D)}\right)} \\ x_p = \sqrt{\dfrac{2\varepsilon_0\varepsilon_r(V_{bi} - V_{app})}{e}\left(\dfrac{N_A}{N_D(N_A+N_D)}\right)} \end{cases}$$

IV Characteristic of PN Junction.



① Depletion region approximation

② non-degenerate semi-conductor $\Rightarrow$ Boltzmen Distribution

③ low-level injection & complete ionization

track minority carrier in Quasi-neutral region.

④ I_{total} is Constant

⑤ no generation and recombination in depletion region $\Rightarrow \begin{cases} I_n = \text{Constant} \\ I_p = \text{Constant} \end{cases}$

$x \ll -x_p$, $J_n = en\mu_n E + eD_n \frac{\partial n}{\partial x}$

$E \approx 0$

$\frac{\partial n}{\partial t} = \frac{1}{e}\cdot\frac{\partial J_n}{\partial x} + g_n - r_n$

steady state: $\frac{\partial n}{\partial t} = 0$

$\Rightarrow D_n \frac{\partial^2 n}{\partial x^2} + g_n - r_n = 0$. $n(x) = n_0 + \delta n(x)$, $g_n - r_n = -\frac{\delta n(x)}{\tau_n}$

$\frac{\partial^2 n_0}{\partial x^2} = 0$.

$D_n \frac{\partial^2 \delta n}{\partial x^2} - \frac{\delta n}{\tau_n} = 0 \Longrightarrow \frac{\partial^2 \delta n}{\partial x^2} - \frac{\delta n}{D_n \tau_n} = 0$

assume $L_n^2 = D_n \tau_n$. L_n: diffusion length.

$\Rightarrow \frac{\partial^2 \delta n}{\partial x^2} - \frac{\delta n}{L_n^2} = 0$ $\quad x \le -x_p$

$\delta n(x) = A e^{\frac{x}{L_n}} + B e^{-\frac{x}{L_n}} \xrightarrow{x \to -\infty,\ \delta n \to 0} \delta n = A e^{\frac{x}{L_n}} = \delta n(-x_p)\cdot e^{(x+x_p)/L_n}$

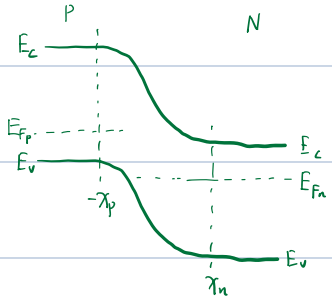
$\delta n(-x_p) = n(-x_p) - n_0(-x_p)$

$n_0(-x_p) = \frac{n_i^2}{p(-x_p)} = \frac{n_i^2}{N_A}$

$J_n = n\mu_n \cdot \frac{\partial E_{Fn}}{\partial x}$

① $x > x_n$. $\frac{\partial E_{Fn}}{\partial x} = 0$. E_{Fn} is flat

② $-x_p \le x < x_n$ $\frac{\partial E_{Fn}}{\partial x} \approx 0$.



E_{Fn} is flat, $x > -x_p$

$n(-x_p) = N_c \cdot \exp\left(\frac{E_{Fn}(-x_p) - E_c(-x_p)}{kT}\right)$ $\quad n_0(-x_p) = N_c \cdot \exp\left(\frac{E_{Fn}^{(0)}(-x_p) - E_c^{(0)}(-x_p)}{kT}\right)$

E_{Fn} is flat $\Rightarrow \frac{n(-x_p)}{n_0(-x_p)} = \exp\left(\frac{E_c^{(0)}(-x_p) - E_c(-x_p)}{kT}\right) = \exp\left(\frac{eV_{app}}{kT}\right)$

$\delta n(-x_p) = n_0(-x_p)\cdot\left[\exp\left(\frac{eV_{app}}{kT}\right) - 1\right]$

$\delta n(-x_p) = \frac{n_i^2}{N_A}\cdot\left[\exp\left(\frac{eV_{app}}{kT}\right) - 1\right]$

$x < -x_p$

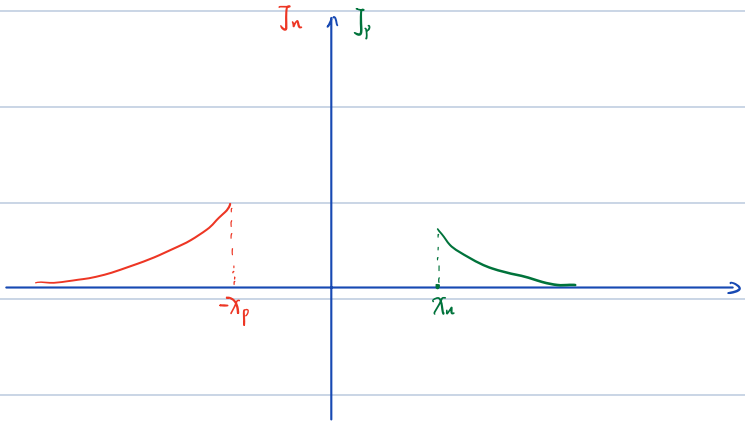
$$\delta n(x) = \frac{n_i^2}{N_A}\cdot\left[\exp\left(\frac{eV_{app}}{kT}\right) - 1\right]\cdot\exp\left(\frac{x + x_p}{L_p}\right)$$

$$J_n(x) = eD_n\frac{\partial n}{\partial x} = eD_n\frac{\partial \delta n}{\partial x} = \frac{eD_n}{L_n}\cdot\frac{n_i^2}{N_A}\cdot\left[\exp\left(\frac{eV_{app}}{kT}\right) - 1\right]\exp\left(\frac{x + x_p}{L_p}\right)$$

$x > x_n$

$$\delta p(x) = \frac{n_i^2}{N_D}\cdot\left[\exp\left(\frac{eV_{app}}{kT}\right) - 1\right]\cdot\exp\left(-\frac{(x - x_n)}{L_p}\right)$$

$$J_n(x) = eD_n\frac{\partial n}{\partial x} = eD_n\frac{\partial \delta n}{\partial x} = \frac{eD_n}{L_n}\cdot\frac{n_i^2}{N_A}\cdot\left[\exp\left(\frac{eV_{app}}{kT}\right) - 1\right]\exp\left(-\frac{(x - x_n)}{L_p}\right)$$

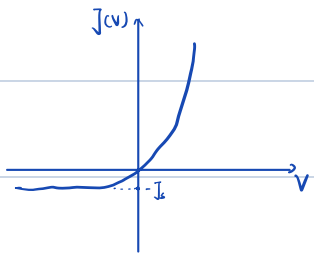


⑤ no generation and recombination in depletion region $\Rightarrow \begin{cases} I_n = \text{Constant} \\ I_p = \text{Constant} \end{cases}$

$$J(x) = J_p(x) + J_n(x) = J_p(x_n) + J_n(-x_p)$$

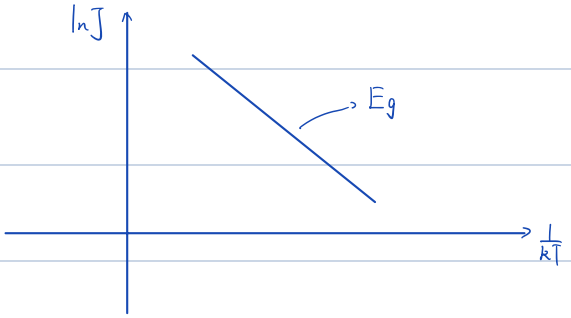
$$J(x) = \left[\frac{eD_p}{L_p}\cdot\frac{n_i^2}{N_D} + \frac{eD_n}{L_n}\cdot\frac{n_i^2}{N_A}\right]\left[\exp\left(\frac{eV_{pn}}{kT}\right) - 1\right]$$

$$\Rightarrow J(V) = J_s\left(e^{\frac{eV}{kT}} - 1\right)$$

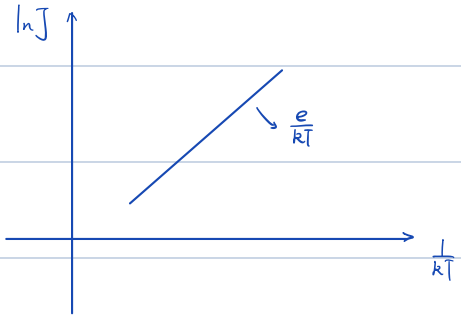


$V < 0$, $|V| \gg \frac{kT}{e}$ $\quad J(V) = -\left[\frac{eD_p}{L_p}\cdot\frac{n_i^2}{N_D} + \frac{eD_n}{L_n}\cdot\frac{n_i^2}{N_A}\right] \propto n_i^2$ (diffusion dominate current) $\quad n_i^2 \propto \exp\left(-\frac{E_g}{2kT}\right)$

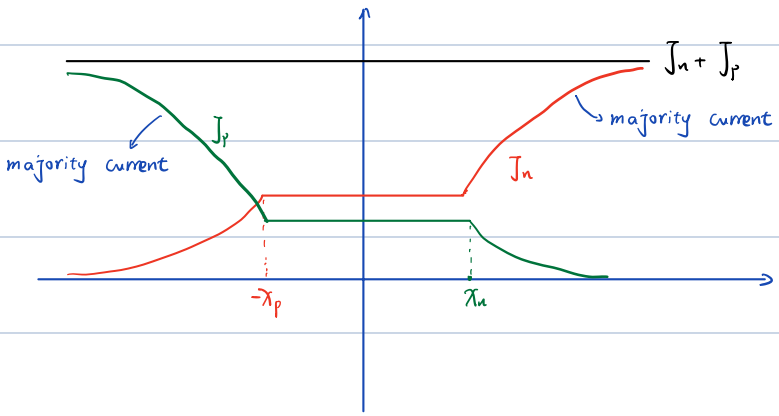
$$\Rightarrow \ln J \propto -\frac{E_g}{kT}$$



$V > 0$. $\quad J(V) = J_s \cdot \exp\left(\frac{eV}{kT}\right)$

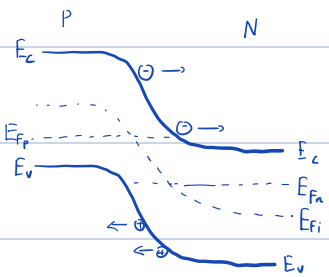


no G & R $\Rightarrow$ in depletion region. J_n = Const. J_p = Const.



with G & R. $R_{net} = \frac{np - n_i^2}{\tau_p(n+n_i) + \tau_n(p+p_i)}$

① $V < 0$



$n = n_i \cdot \exp\left(\frac{E_{Fn} - E_{Fi}}{kT}\right) \quad p = n_i \cdot \exp\left(\frac{E_{Fi} - E_{Fp}}{kT}\right)$

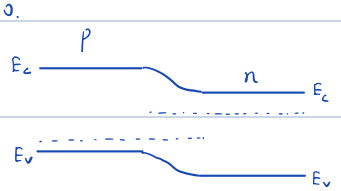
$np = n_i^2 \cdot \exp\left(\frac{E_{Fn} - E_{Fp}}{kT}\right) = n_i^2 \cdot \exp\left(\frac{eV}{kT}\right) < n_i^2$

$n, p \longrightarrow 0 \quad R_{net} = \frac{-n_i}{\tau_p + \tau_n} = -G_{net}$

$\frac{\partial n}{\partial t} = \frac{1}{e} \cdot \frac{\partial J_n}{\partial x} + G_{net} = 0 \quad -x_p < x < x_n \quad J_n = e\int_{-x_p}^{x_n} (-G_{net})\,dx = -e \cdot \frac{n_i}{\tau_p + \tau_n} \cdot W \propto n_i$

② $V > 0$.

$\tau_n = \tau_p = \tau_0, \quad n_i = p_i = n_i$



$R_{net} = \frac{np - n_i^2}{\tau_p(n+n_i) + \tau_n(p+p_i)} = \frac{n_i^2\left(\exp\left(\frac{eV}{kT}\right) - 1\right)}{2\tau_0 n_i\left(\exp\left(\frac{eV}{2kT}\right) + 1\right)} \xrightarrow{\frac{eV}{kT} \gg 1} R_{net,max} = \frac{n_i}{2\tau_0}\exp\left(\frac{eV}{2kT}\right)$

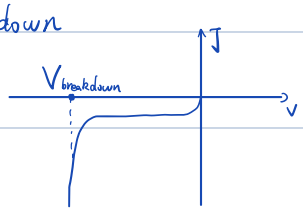
$J_{G\&R} = e\int R_{net,max}\,dx = ex\frac{n_i}{2\tau_0}\exp\left(\frac{eV}{2kT}\right) = eW\frac{n_i}{2\tau_0}\cdot\exp\left(\frac{eV}{2kT}\right)$

$$J_{GRR}(V) = J_{r_0}\left[\exp\left(\frac{eV}{2kT}\right) - 1\right]$$

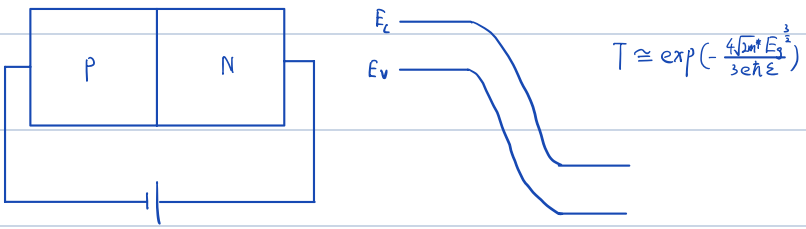
$$J_{diff}(V) = J_s\left[\exp\left(\frac{eV}{kT}\right) - 1\right]$$

$$\Rightarrow J_{total} = J_{GRR}(V) + J_{diff}(V) = J_0\left[\exp\left(\frac{eV}{nkT}\right) - 1\right]$$ n: ideality factor $\begin{cases} n=2 \rightarrow J_{GRR} \\ n=1 \rightarrow J_{diff} \end{cases}$

Breakdown



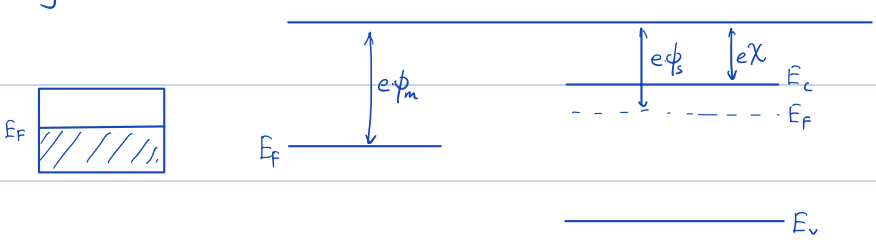
① tunneling → Zener breakdown



$$T \cong \exp\left(-\frac{4\sqrt{2m^*}E_g^{\frac{3}{2}}}{3e\hbar\mathcal{E}}\right)$$

② Impact ionization → avalanche breakdown

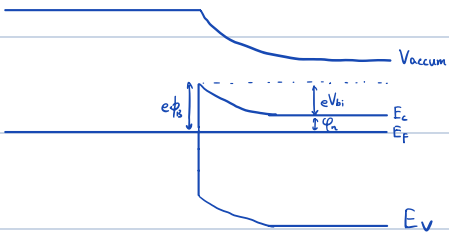
Schottky Diodes



ϕ_m: work function of metal

ϕ_s: work function of Semiconductor

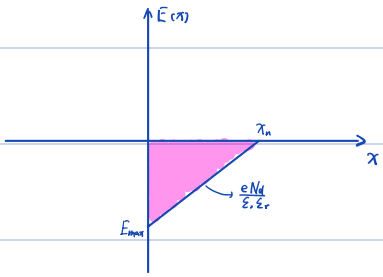
χ: Affinity



$\phi_B = \phi_m - \chi$

$\varphi_n = \phi_s - \chi$

$V_{bi} = \phi_B - \varphi_n = \phi_m - \phi_s = \phi_{ms}$



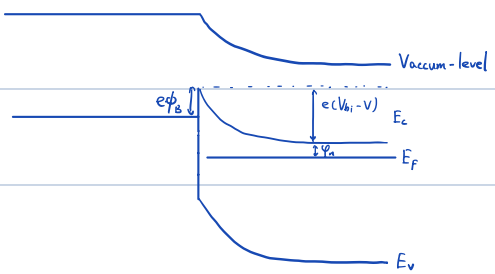
$E_{max} = x_n \cdot \frac{eN_d}{\varepsilon_s\varepsilon_r}$

$V_{bi} = \frac{1}{2} E_{max} \cdot x_n = \frac{eN_d}{2\varepsilon_s\varepsilon_r} \cdot x_n^2$

$W = x_n = \sqrt{\frac{2V_{bi}\cdot\varepsilon_s\varepsilon_r}{eN_d}} \quad \xrightarrow{V_{bi} \rightarrow V_{bi}-V} \quad W(V) = \sqrt{\frac{2(V_{bi}-V)\varepsilon_s\varepsilon_r}{eN_d}}$

$C_j(V) = \frac{A\cdot\varepsilon_s\varepsilon_r}{W} = A\sqrt{\frac{eN_d\cdot\varepsilon_s\varepsilon_r}{2(V_{bi}-V)}}$

$V < 0$



$J(V) = J_{s\to m}(V) - J_{m\to s}(V)$, $V=0$: $J_{(0)} = J_{s\to m}(0) - J_{m\to s}(0) = 0$

$J_{m\to s}(V) = J_{m\to s}(0)$, $J_{s\to m}(V) = J_{s\to m}(0)\cdot e^{\frac{eV}{kT}}$

$J(V) = J_{s\to m}(V) - J_{m\to s}(V) = J_{s\to m}(0)e^{\frac{eV}{kT}} - J_{m\to s}(0)$

$J(V) = J_{s\to m}(0)\left[e^{\frac{eV}{kT}} - 1\right]$, $J_{s\to m}(0) = \frac{4\pi m^* e (kT)^2}{h^3} \cdot e^{-(e\phi_s)/kT}$

$$\boxed{J(V) = \frac{4\pi m^* e (kT)^2}{h^3} \cdot e^{\frac{-(e\phi_s)}{kT}}\left[e^{\frac{eV}{kT}} - 1\right]}$$

$$J_{(V)}_{schotty} = \frac{4\pi m^* e (kT)^2}{h^3} \cdot e^{-\frac{(e\phi_B)}{kT}} \left[e^{\frac{eV}{kT}} - 1\right]$$

$$= J_s \left[e^{\frac{eV}{kT}} - 1\right]$$

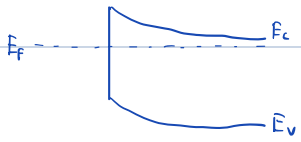
$$J_{(V)}_{PN} = \left[\frac{eD_p}{L_p} \cdot \frac{n_i^2}{N_D} + \frac{eD_n}{L_n} \cdot \frac{n_i^2}{N_A}\right]\left[\exp\left(\frac{eV}{kT}\right) - 1\right]$$

$$= J_{PN} \left[e^{\frac{eV}{kT}} - 1\right]$$

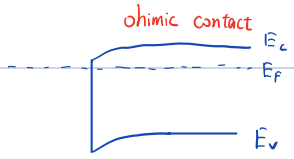
⊙ $J_s >> J_{PN}$ $\quad J_s \propto \exp\left(-\frac{e\phi_B}{kT}\right)$ $\quad J_{PN} \propto n_i^2 \propto \exp\left(-\frac{E_g}{kT}\right)$

⊙ schotty : majority carrier device. $\quad$ PN : minority carrier device

$\phi_m > \phi_s$

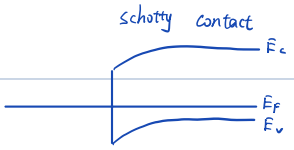


$\phi_m < \phi_s$

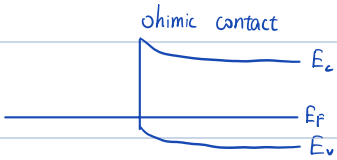


Metal + p-type

$\phi_m < \phi_s$



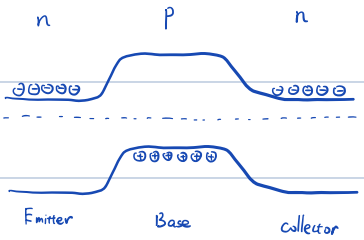
$\phi_m > \phi_s$

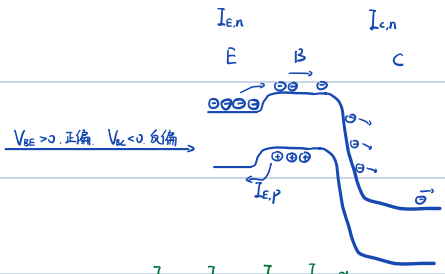


Bipolar Junction Transistors

PNP | P | N | P |

NPN | N | P | N |





$I_{c,n} \propto I_{E,n}$, $I_{c,n} = I_{E,n}\alpha_n$

α_n: base transport factor

$I_E = I_{E,p} + I_{E,n}$

$I_{E,n} = \gamma_e \cdot I_E$. $\gamma_e = \frac{I_{E,n}}{I_E} = \frac{I_{E,n}}{I_{E,n}+I_{E,p}}$.

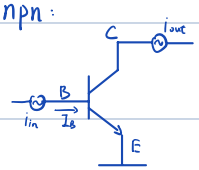
γ_e: emitter injection efficiency.

$I_c = I_{c,n} + I_{c,p} = \alpha_n I_{E,n} + I_{cp}$

$I_c = \alpha_n \cdot \gamma_e \cdot I_E + I_{c,p}$

$\begin{cases} I_{c,p} \text{ is small} \\ I_E \longrightarrow \delta I_E \\ I_c \longrightarrow \delta I_c \end{cases} \longrightarrow \alpha = \frac{\delta I_c}{\delta I_e} = \alpha_n \gamma_e$ ⟶ Common base current gain

npn:



$I_c = \alpha I_E + I_{cp}$

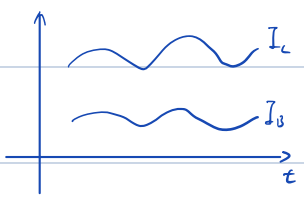
$I_E = I_c + I_B$

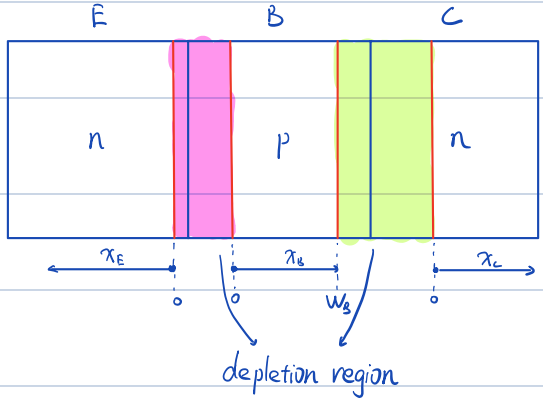
$I_c = \alpha(I_c + I_B) + I_{cp}$

$I_c = \frac{\alpha}{1-\alpha} I_B + \frac{I_{cp}}{1-\alpha}$

$\beta = \frac{\delta I_c}{\delta I_B} = \frac{\alpha}{1-\alpha}$ ⟶ Common emitter current gain

$\beta \gg 1$. input I_B, output I_c



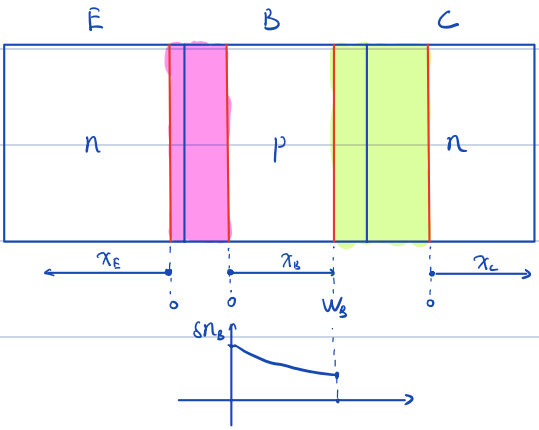


$V_{BE} > 0, \quad V_{BC} < 0$

$n_B(x_B=0) = n_{B_0} e^{eV_{BE}/kT}, \quad \delta n_B\big|_{x_B=0} = n_B - n_{B_0} = n_{B_0}(e^{eV_{BE}/kT} - 1), \quad n_{B_0} = \frac{n_{B,i}^2}{N_{B,A}}$

$\delta n_B\big|_{x_B=W_B} = n_B(e^{eV_{BE}/kT} - 1)$

$D_{B,n}\frac{\partial^2 \delta n_B}{\partial x_B^2} - \frac{\delta n_B}{\tau_{B,n}} = 0 \Rightarrow \delta n_B(x_B) = A\cdot e^{-x_B/L_{B,n}} + B\cdot e^{x_B/L_{B,n}}, \quad L_{B,n} = \sqrt{D_{B,n}\tau_{B,n}}$



$\delta n_B(0) = A + B$

$\delta n_B(W_B) = A e^{-W_B/L_{B,n}} + B e^{W_B/L_{B,n}}$

$\Rightarrow \delta n_B(x_B) = \delta n_B(0)\cdot \frac{\sinh\left(\frac{W_B - x_B}{L_{B,n}}\right)}{\sinh\left(\frac{W_B}{L_{B,n}}\right)} + \delta n_B(W_B)\cdot \frac{\sinh\left(\frac{x_B}{L_{B,n}}\right)}{\sinh\left(\frac{W_B}{L_{B,n}}\right)}$

$I_{E,n} = eAD_{Bn}\cdot \frac{\partial \delta n_B}{\partial x_B}\Big|_{x_B=0}, \quad I_{C,n} = eAD_{Bn}\cdot \frac{\partial \delta n_B}{\partial x_B}\Big|_{x_B=W_B}, \quad \alpha_n = \frac{I_{Cn}}{I_{E,n}}$

$x \ll 1 \longrightarrow \sinh x \approx x \qquad W_B \ll L_{B,n}. \qquad \delta n_B(x_B) = \delta n_B(0)\cdot \frac{\frac{W_B - x_B}{L_{B,n}}}{\frac{W_B}{L_{B,n}}} + \delta n_B(W_B)\cdot \frac{\frac{x_B}{L_{B,n}}}{\frac{W_B}{L_{B,n}}}$

$\delta n_B(x_B) = \delta n_B(0)\cdot \frac{W_B - x_B}{W_B} + \delta n_B(W_B)\cdot \frac{x_B}{W_B}$

$I_{E,n} \cong I_{C,n}, \quad \alpha_n = 1$

$I_{E,n} = eAD_{Bn}\frac{\partial \delta n_B}{\partial x_B}\Big|_{x_B=0} = eAD_{Bn}\left[\frac{\delta n_B(0)}{\sinh\left(\frac{W_B}{L_B}\right)}\cdot \cosh\left(\frac{W_B}{L_{Bn}}\right)\cdot\left(-\frac{1}{L_{Bn}}\right) + \frac{\delta n_B(W_B)}{\sinh\left(\frac{W_B}{L_{Bn}}\right)}\cdot\left(\frac{1}{L_{Bn}}\right)\right]$

$I_{C,n} = eAD_{Bn}\frac{\partial \delta n_B}{\partial x_B}\Big|_{x_B=W_B} = eAD_{Bn}\left[\frac{\delta n_B(0)}{\sinh\left(\frac{W_B}{L_B}\right)}\cdot\left(-\frac{1}{L_{Bn}}\right) + \frac{\delta n_B(W_B)}{\sinh\left(\frac{W_B}{L_{Bn}}\right)}\cdot\frac{\cosh\left(\frac{W_B}{L_{Bn}}\right)}{L_{B,n}}\right]$

$\alpha_n = \frac{I_{Cn}}{I_{Bn}} = \frac{\delta n_B(0)\cdot(-1) + \delta n_B(W_B)\cdot\cosh\left(\frac{W_B}{L_{Bn}}\right)}{\delta n_B(0)\cdot\cosh\left(\frac{W_B}{L_{Bn}}\right)\cdot(-1) + \delta n_B(W_B)}$

$\begin{cases} \delta n_B(0) = n_{B_0}(e^{eV_{BE}/kT} - 1) \\ \delta n_B(W_B) = n_{B_0}(e^{eV_{BE}/kT} - 1) \end{cases}$

$\alpha_n = \frac{-(e^{eV_{BE}/kT} - 1) - \cosh\left(\frac{W_B}{L_{Bn}}\right)}{-(e^{eV_{BE}/kT} - 1)\cosh\left(\frac{W_B}{L_{Bn}}\right) - 1} \xrightarrow{e^{eV_{BE}/kT} \gg 1} \frac{1}{\cosh\left(\frac{W_B}{L_{Bn}}\right)} = \frac{1}{1 + \frac{1}{2}\left(\frac{W_B}{L_{Bn}}\right)^2} = 1 - \frac{1}{2}\left(\frac{W_B}{L_{Bn}}\right)^2$

$$\gamma_e = \frac{I_{E,n}}{I_E} = \frac{I_{E,n}}{I_{E,n} + I_{E,p}} \quad . \quad I_{E,p} = -eD_{E,p} \cdot \frac{\partial \delta P_E}{\partial x_E}\Big|_{x_E=0} \quad . \quad D_{E,p} \cdot \frac{\partial^2 P_E}{\partial x_E^2} - \frac{\delta P_E}{\tau_{E,p}} = 0$$

$$\delta P_E = A e^{-\frac{x_E}{L_p}} + B' e^{\frac{x_E}{L_{E,p}}} \quad . \quad x_E \longrightarrow \infty \; . \; \delta P_E \longrightarrow 0 \Rightarrow \delta P_E = \delta P_E(x_E=0) \cdot e^{-x_E/L_{E,p}} \quad \vdots \quad \delta P_E(x_E=0) = P_{E_0}\left(e^{\frac{eV_{BE}}{kT}} - 1\right)$$

$$I_{E,p} = +eAD_{E,p} \cdot \frac{\partial \delta P_E}{\partial x_E}\Big|_{x_E=0} = -\frac{eAD_{E,p}}{L_{E,p}} \cdot P_{E_0}\left(e^{eV_{BE}/kT} - 1\right)$$

$$I_{E,n} \cong eAD_{Bn}\left[\frac{\delta n_B(w)}{\sinh\left(\frac{W_B}{L_B}\right)} \cdot \cosh\left(\frac{W_B}{L_{Bn}}\right) \cdot \left(-\frac{1}{L_{Bn}}\right)\right]$$

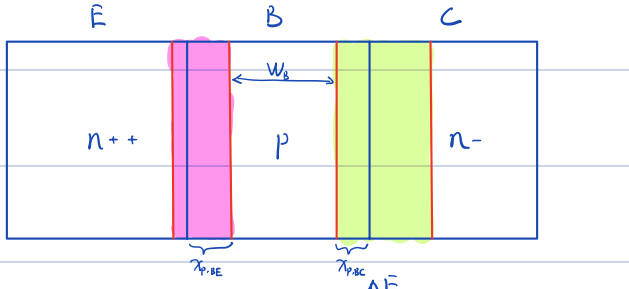
$$\gamma_e = \frac{I_{En}}{I_{En} + I_{Ep}} = \frac{1}{1 + \frac{D_{E,p} P_{E_0} L_{Bn} \sinh\left(\frac{W_B}{L_{Bn}}\right)}{D_{Bn} n_{B_0} L_{Ep} \cosh\left(\frac{W_B}{L_{Bn}}\right)}} \xrightarrow{W_B \ll L_{Bn}} \frac{1}{1 + \frac{D_{Ep}}{D_{Bn}} \frac{P_{E_0}}{n_{B_0}} \frac{W_B}{L_{Ep}}}$$

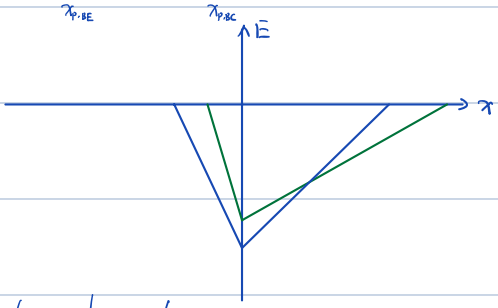
$$\Rightarrow \boxed{\alpha = \gamma_e \alpha_n = \frac{1}{1 + \frac{D_{Ep}}{D_{Bn}} \frac{P_{E_0}}{n_{B_0}} \frac{W_B}{L_{Ep}}} \cdot \frac{1}{\cosh\left(\frac{W_B}{L_{Bn}}\right)}}$$

$$\beta = \frac{\alpha}{1-\alpha} \cong \frac{D_{Bn}}{D_{Ep}} \cdot \frac{n_{B_0}}{P_{E_0}} \cdot \frac{L_{Ep}}{W_B} \quad \vdots \quad n_{B_0} = \frac{n_{iB}^2}{N_{B,A}} \; . \; P_{E_0} = \frac{n_{iE}^2}{N_{E,D}}$$

$$\Rightarrow \boxed{\beta = \frac{D_{Bn} \cdot n_{iB}^2 \cdot N_{E_D} \cdot L_{Ep}}{D_{Ep} \cdot n_{iE}^2 \cdot N_{B_A} \cdot W_B}}$$

Design of BJT





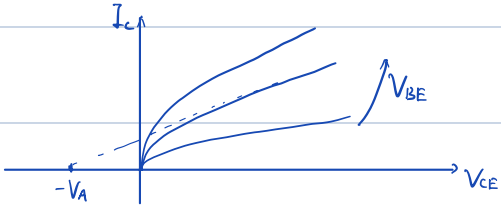
β: $N_E > N_B$, $N_B > N_C$

$N_{B,A}$ ↓ β ↗.

Problem of low base doping

① Base width modulation. $W_B = W_{B,total} - x_{p,BE} - x_{p,BC}$

$|V_{BC}|$ ↗ $x_{p,BC}$ ↗. W_B ↓. β ↗. ⇒ β depends on V_{BC} (V_{CE}) Bad

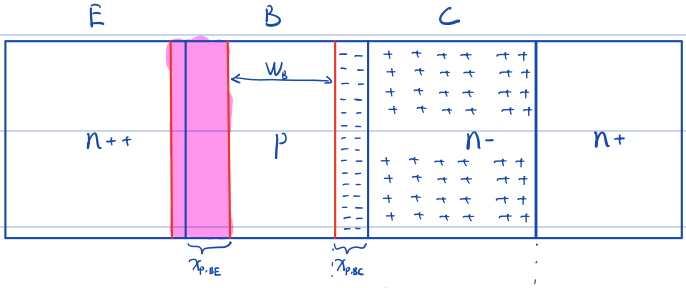


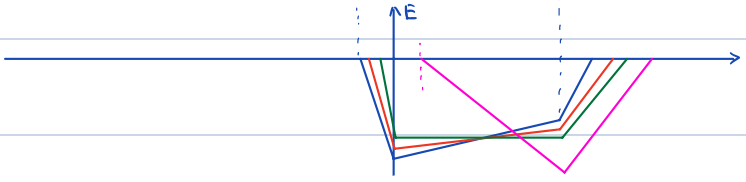
V_A: early voltage

$$\frac{\delta I_C}{\delta V_{CE}} = \frac{I_C}{V_{CE} + V_A}$$

② Current crowding

solution: reduce collector doping $N_{C,D}$. issue: kirk effect

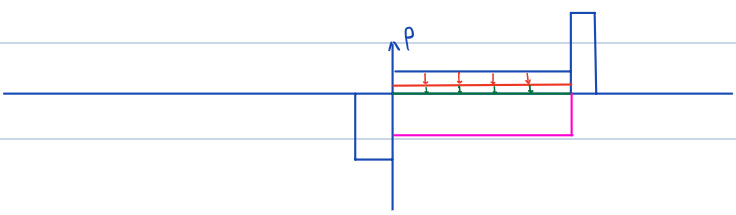




$J = J_{critical}$

$J > J_{critical}$. $J = J_{kirk}$, 部分 collector 变成 Base

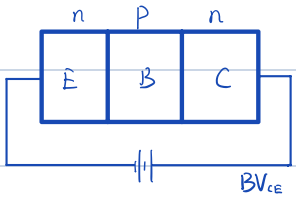
Base push out. W_B ↗. β ↓

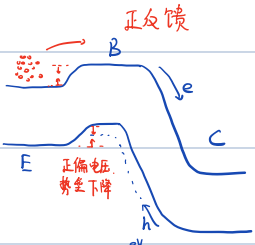


$$\beta = \frac{D_{Bn} \cdot n_{iB}^2 \cdot N_{E,D} \cdot L_{EP}}{D_{EP} \cdot n_{iE}^2 \cdot N_{B,A} \cdot W_B}$$

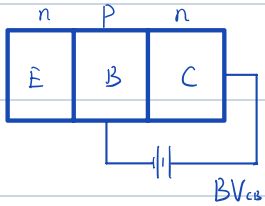
Increase emitter doping. $N_{E,D}$ ↗. E_g ↓. n_i ↗. β ↓

Breakdown Voltage





$I \propto e^{\frac{eV}{kT}} - 1$



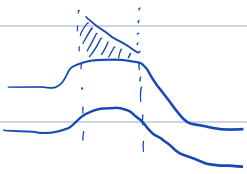
$|BV_{CE}| < |BV_{CB}|$

Frequency response

$\tau_{EC} = \tau_{BE} + \tau_B + \tau_{BC}$. $f = \frac{1}{2\pi \tau_{EC}}$

① $\tau_{BE} = \frac{\Delta Q_{BE}}{\Delta I_E} = \frac{C_{BE} \cdot \Delta V_{BE}}{\Delta I_E} = C_{BE} \frac{1}{\frac{\partial I_E}{\partial V_{BE}}} = \frac{C_{BE} \cdot kT}{e \cdot I_E} \approx C_{BE} \frac{kT}{eI_c}$

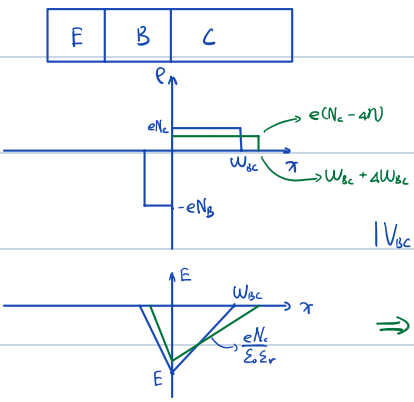
$I_E \propto \exp(\frac{eV_{BE}}{kT})$. $I_c \uparrow \tau_{BE} \downarrow f \uparrow$



② $\tau_B = \frac{\Delta Q}{\Delta I} = \frac{q \cdot n \cdot W_B \cdot \frac{1}{2}}{q \cdot D_n \cdot \frac{n}{W_B}} = \frac{W_B^2}{2D_n} \propto W_B^2$

↳ diffusion time

③ $\tau_{BC} = \frac{W_{BC}}{v_s} \Rightarrow \tau_{BC} = \frac{W_{BC}}{2v_s}$



$|V_{BC}| = \frac{E W_{BC}}{2} = \frac{eN_c W_{BC}^2}{2\varepsilon_0 \varepsilon_r}$

$\Rightarrow |V_{BC}| = \frac{e(N_c - \Delta n)(W_{BC} + \Delta W_{BC})^2}{2\varepsilon_0 \varepsilon_r}$

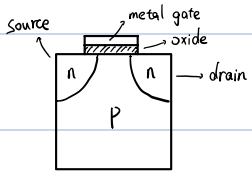
$N_c W_{BC}^2 = (N_c - \Delta n)(W_{BC} + \Delta W_{BC})^2 \Rightarrow \boxed{\Delta n \, W_{BC} = 2N_c \, \Delta W_{BC}}$

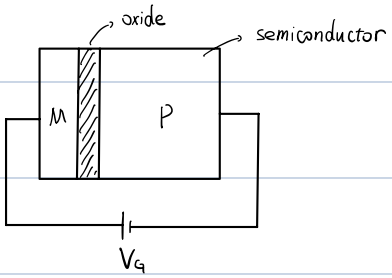
$\tau_{BC} = \frac{\Delta Q_{BC}}{\Delta I_{BC}}$, $\Delta Q_{BC} = q(\Delta n \cdot W_{BC} - (N_c - \Delta n)\Delta W_{BC}) \cong q(\Delta n \cdot W_{BC} - N_c \cdot \Delta W_B) = \frac{q \Delta n \cdot W_{BC}}{2}$

$\Delta I_{BC} = q \cdot \Delta n \cdot v_s \cdot A$

$\Rightarrow \tau_{BC} = \frac{W_{BC}}{2v_s}$

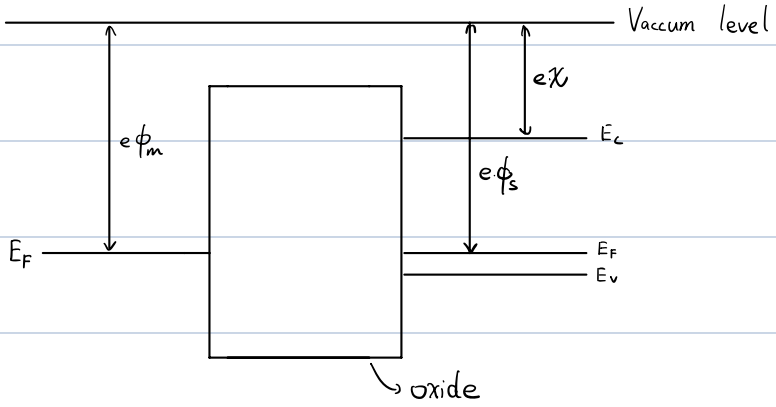
MOSFET



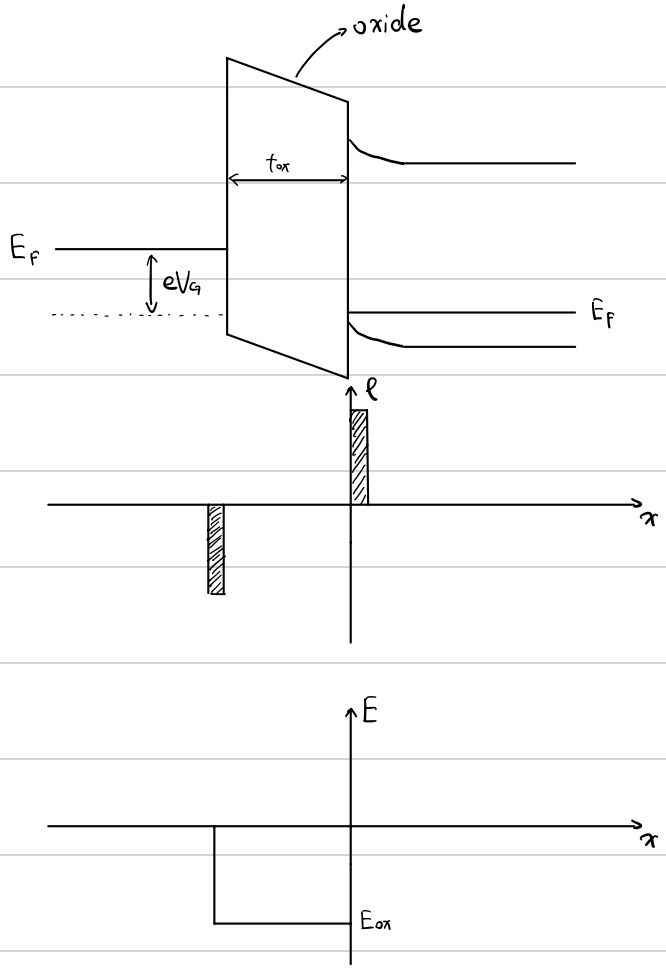


assume $\phi_m = \phi_s$

$V_G = 0$

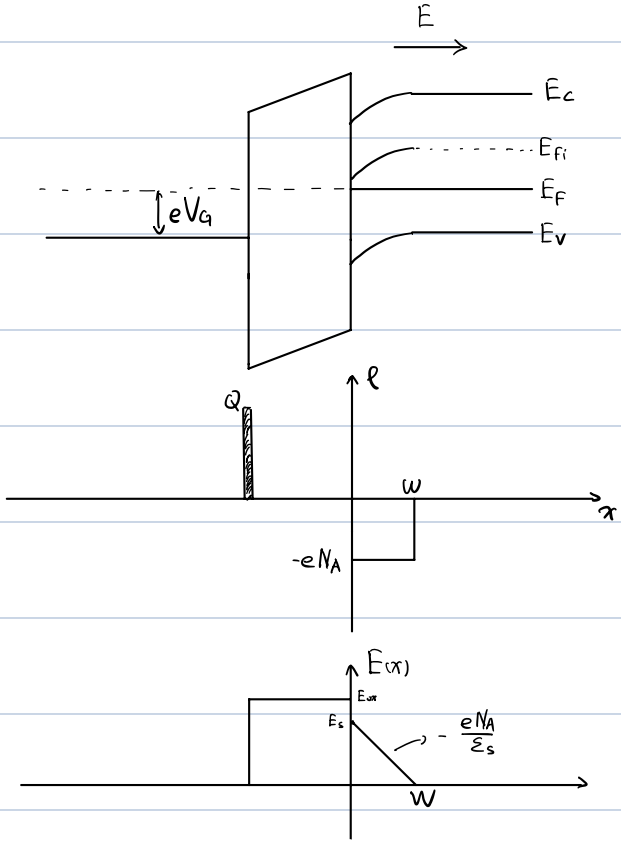


$V_G < 0$

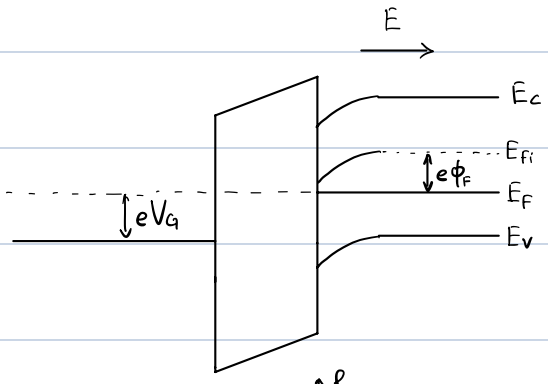


$V_G = E_{ox} \cdot t_{ox}$

$V_G > 0$



介电常数不一样. D(x)连续. E(x)不连续.

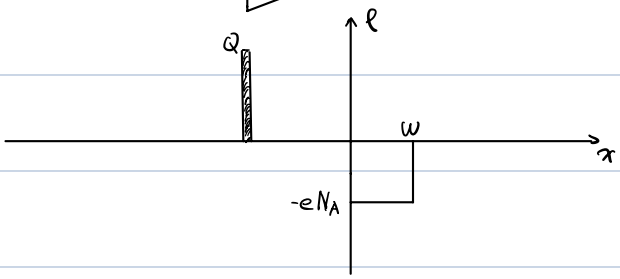


$$p = n_i \exp\left(\frac{E_{Fi} - E_F}{kT}\right) = n_i \exp\left(\frac{e\phi_F}{kT}\right)$$

$$\varphi_F = \frac{kT}{e} \ln \frac{N_A}{n_i}$$

$E_{Fi}^{(s)} = E_{Fi}^{(b)} - 2\phi_F$

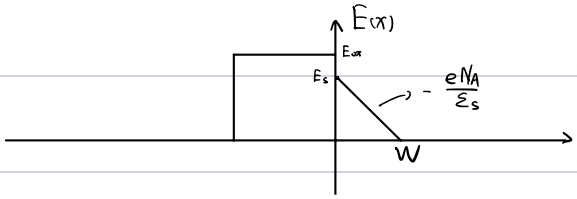
$$n_s = n_i \exp\left(\frac{E_F - E_{Fi}^{(s)}}{kT}\right) = n_i \exp\left(\frac{E_F - E_{Fi}^{(b)} + 2e\phi_F}{kT}\right) = n_i \exp\left(\frac{e\phi_F}{kT}\right) = N_A$$



⇒ 半导体压降: $\psi_s = 2\phi_F$, $\psi_s = \frac{eN_A \cdot w^2}{2\varepsilon_s}$

$$\Rightarrow w = \sqrt{\frac{4\varepsilon_s \phi_F}{eN_A}}$$

$$V_G = V_{ox} + \psi_s$$

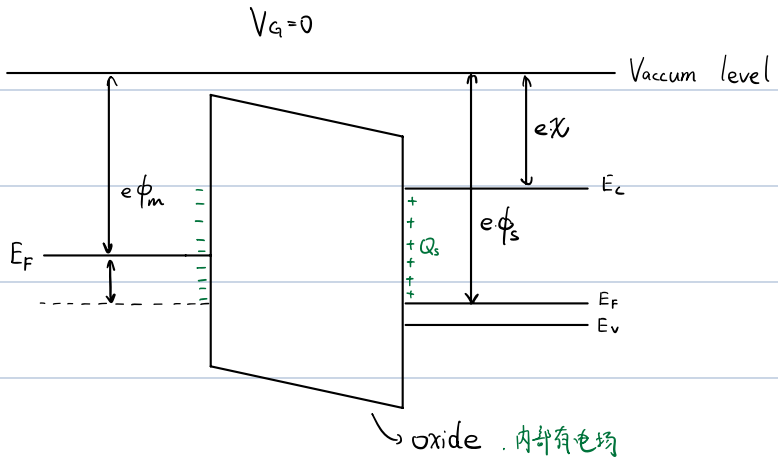


$$= \frac{eN_A \cdot w}{\varepsilon_{ox}/t_{ox}} + 2\phi_F \quad , \quad Q_s = eN_A \cdot w$$

$$V_G = \frac{Q_s}{C_{ox}} + 2\phi_F = \frac{\sqrt{4\varepsilon_s \phi_F eN_A}}{C_{ox}} + 2\phi_F \rightarrow V_{th}$$

$$V_G = \frac{eN_A}{\varepsilon_{ox}} \cdot w \cdot t_{ox} + \frac{eN_A}{\varepsilon_s} \cdot \frac{w^2}{2}$$

Assume $\phi_m < \phi_s$

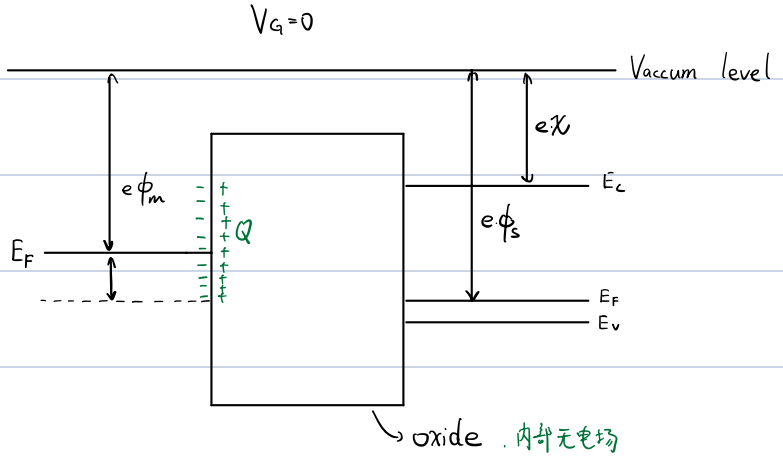


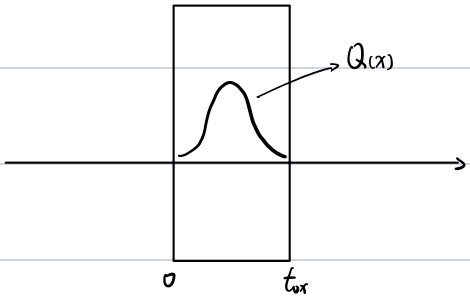
$V_{Flat} < 0.$

$$V_{Flat} = \phi_m - \phi_s = \phi_{ms}$$

$$V_{th}^* = V_{Flat} + V_{th}$$

$$\Delta V_{th} = -\frac{Q_{ss}}{C_{ox}}$$





$$\frac{dE}{dx} = \frac{Q(x)}{\epsilon_{ox}} \qquad E(x)|_{x=t_{ox}} = 0$$

$$E(x) = \int_{t_{ox}}^{x} \frac{Q(x)}{\epsilon_{ox}} dx + E(t_{ox}) = -\int_{x}^{t_{ox}} \frac{Q(x)}{\epsilon_{ox}} dx$$

$$V(0) - V(t_{ox}) = \int_0^{t_{ox}} E(x)\, dx = -\int_0^{t_{ox}} \int_x^{t_{ox}} \frac{Q(x')}{\epsilon_{ox}} dx'\, dx$$

$$\Rightarrow \Delta V_{th} = V(0) - V(t_{ox}) = -\int_0^{t_{ox}} x \cdot \left(\frac{Q(x)}{\epsilon_{ox}}\right) dx$$

$$V_{th}^* = V_{th} + \underbrace{\phi_{ms} + \left(-\frac{Q_{ss}}{C_{ox}}\right) - \int_0^{t_{ox}} x \cdot \frac{Q(x)}{\epsilon_{ox}} dx}_{V_{Flat}}$$

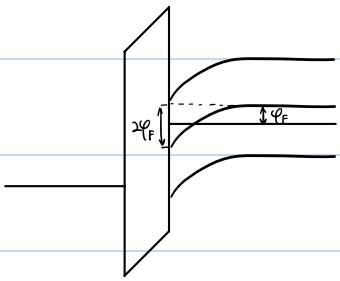
ϕ_{ms}：半导体与金属功函数不同

$-\frac{Q_{ss}}{C_{ox}}$：半导体界面处有电荷

$\int_0^{t_{ox}} x \cdot \frac{Q(x)}{\epsilon_{ox}} dx$：oxide 内部有电荷

① $V < V_{Flat}$. accumulation. 空穴累积.

② $V_{Flat} < V < V_{th}^*$. depletion. 形成耗尽区



③ $V > V_{th}^*$. inversion. $V = V_{ox} + \psi_s + V_{Flat}$.

$V = V_{th}^* = V_{ox} + 2\varphi_F + V_{Flat}$. $\psi_s = 2\varphi_F$

$V \uparrow$. assume：ψ_s 线性 $\uparrow$

inversion electron density $n_{inv} \propto e^{\frac{e\psi_s}{kT}}$：指数 $\uparrow$

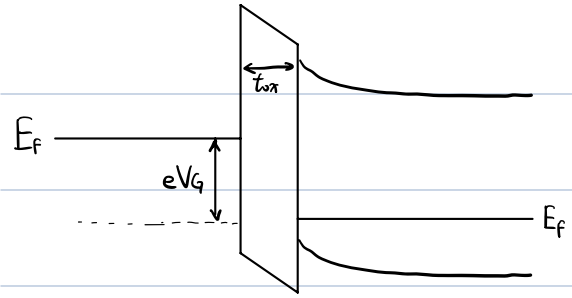
$V_{ox} = \frac{Q_s + n_{inv}}{C_{ox}}$：指数 $\uparrow$

$V = V_{ox} + \psi_s + V_{Flat}$：指数 $\uparrow$ $\Longrightarrow$ 与 V 线性 $\uparrow$ 矛盾.

$\Rightarrow$ when $V > V_{th}^*$ $\psi_s \approx 2\varphi_F$. ψ_s 基本上维持不变

$$\begin{cases} V_{th}^* = V_{ox} + 2\varphi_F + V_{Flat} \\ V = \frac{Q_s + Q_n}{C_{ox}} + 2\varphi_F + V_{Flat}, \quad V > V_{th}^* \end{cases}$$

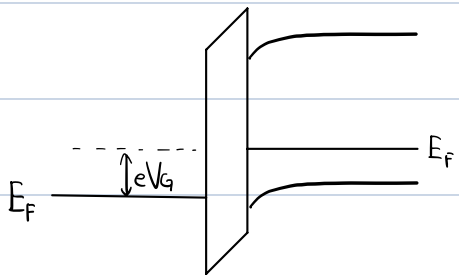
$$\Rightarrow (V - V_{th}^*) = \frac{Q_n}{C_{ox}} \qquad Q_n = C_{ox}(V - V_{th}^*)$$



① $V < V_{Flat}$

$$C = \frac{dQ}{dV} = C_{ox} = \frac{\epsilon_{ox}}{t_{ox}}$$

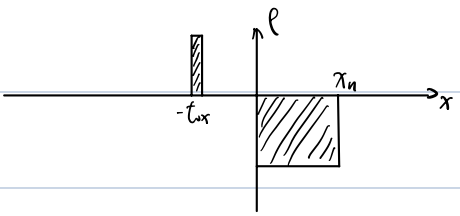
② $V_{Flat} < V < V_{th}^*$

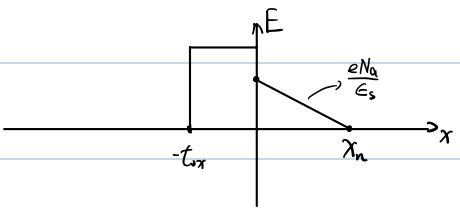


$$C = \frac{dQ}{dV} \qquad dV = dV_{ox} + d\psi_s$$

$$\frac{1}{C} = \frac{dV}{dQ} = \frac{dV_{ox}}{dQ} + \frac{d\psi_s}{dQ}$$

$\frac{dV_{ox}}{dQ} \rightarrow \frac{1}{C_{ox}}$

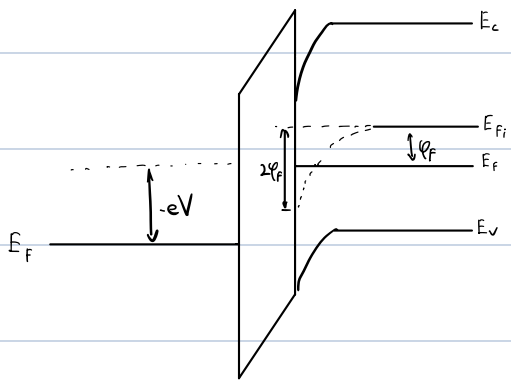




$$\psi_s = \frac{eN_a}{\epsilon_s} x_n^2 \cdot \frac{1}{2} \qquad Q = eN_a x_n \qquad \frac{d\psi_s}{dQ} = \frac{x_n}{\epsilon_s} = \frac{1}{C_s}$$

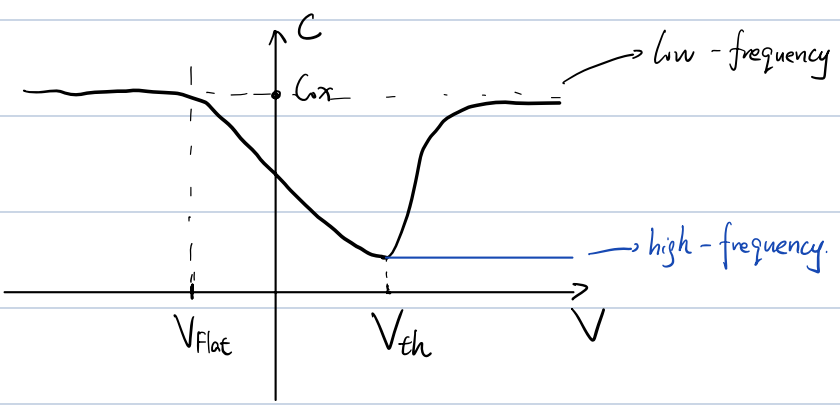
$$\Rightarrow \frac{1}{C} = \frac{1}{C_{ox}} + \frac{1}{C_s}$$

③ $V > V_{th}^*$

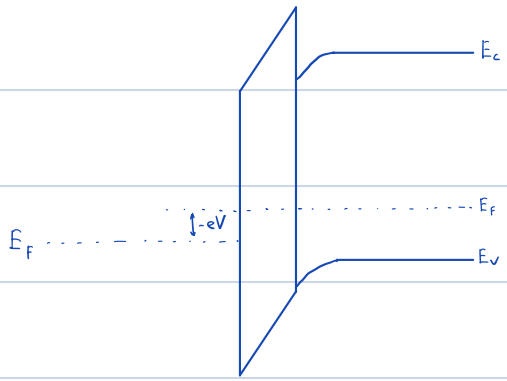


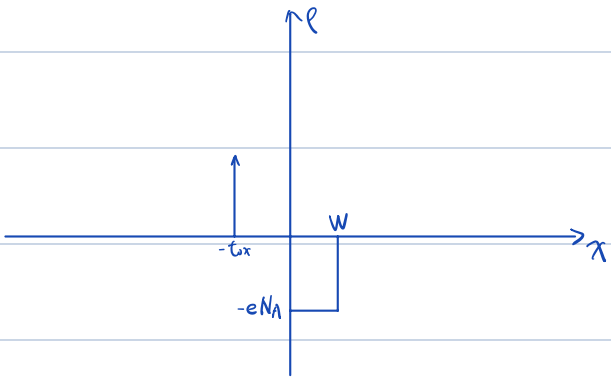
$$\varphi_s \cong 2\varphi_F \qquad dV = dV_{ox} \qquad d\varphi_s = 0$$

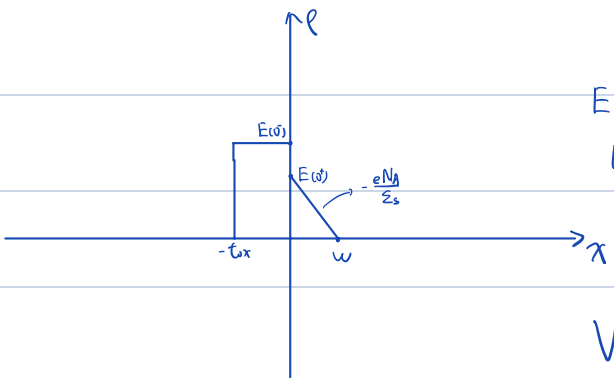
$$C = \frac{dQ}{dV} = \frac{dQ}{dV_{ox}} = C_{ox}$$



$V > 0$



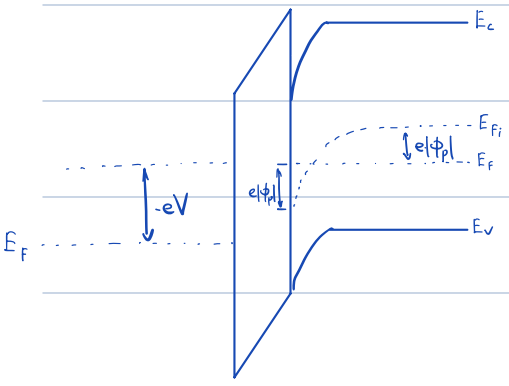




$$E(0^+) = \frac{eN_A}{\varepsilon_s} W$$

$$D(0^-) = D(0^+) \Rightarrow E(0^-) = \frac{eN_A}{\varepsilon_{ox}} \cdot W$$

$$V_G = \int_{-\infty}^{+\infty} E\,dx = V_{ox} + V_s = \frac{eN_A}{\varepsilon_{ox}} \cdot W \cdot t_{ox} + \frac{1}{2} \cdot \frac{eN_A}{\varepsilon_s} \cdot W^2$$



when $V_s = 2|\phi_p| \Rightarrow$ inversion

在表面处的电子浓度与半导体体内的空穴浓度相等.

$$V_G = V_{th} \quad \frac{eN_A}{\varepsilon_s} \cdot \frac{W^2}{2} = 2|\phi_s|$$

$$W = 2\sqrt{\frac{|\phi_p| \cdot \varepsilon_s}{eN_A}} \qquad V_{ox} = \frac{2\sqrt{eN_A \cdot |\phi_p| \varepsilon_s}}{\varepsilon_{ox}} \cdot t_{ox}$$

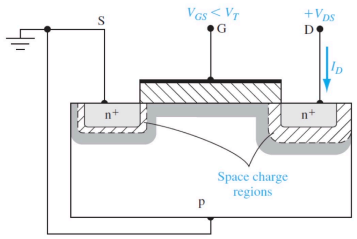
$$\boxed{V_{th} = V_G = \frac{2\sqrt{eN_A \cdot |\phi_p| \varepsilon_s}}{\varepsilon_{ox}} \cdot t_{ox} + 2|\phi_p| = \frac{Q_{ss}}{C_{ox}} + 2|\phi_p|}$$

$Q_{ss} = eN_A \cdot W$

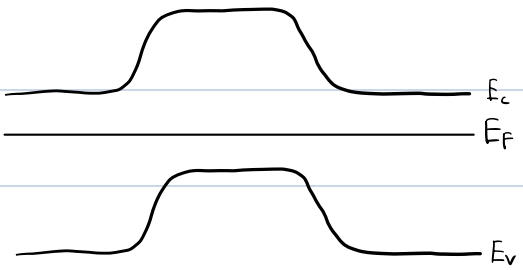
$V_{Flat} = 0$

$$V_{Flat} \neq 0 \quad V_{th} = \frac{Q_{ss}}{C_{ox}} + 2|\phi_p| + V_{Flat}$$

MOSFET



$V_{DS} = 0$



① V_{DS} is small.

$V_G > V_{th}^*$ $\quad |Q_n| = C_{ox}(V_G - V_{th}^*)$ $\quad |Q_n| = q \cdot n \cdot t$

$E = \frac{V_{DS}}{L}$

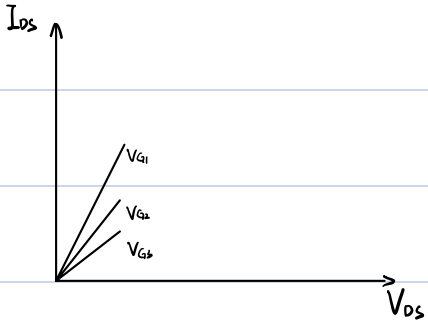
$\begin{cases} A = w \cdot t \\ J_n = q \cdot n \cdot v_n = q \cdot n \cdot \mu_n \cdot E \end{cases}$ $\Rightarrow I_n = J_n \cdot A = q \cdot n \cdot \mu_n \cdot E \cdot w \cdot t$

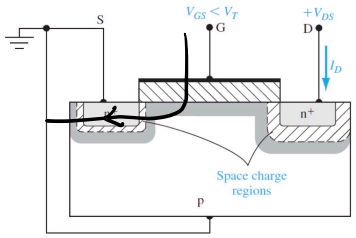
$\Rightarrow I_n = |Q_n| \cdot \mu_n \cdot E \cdot w$

$I_n = |Q_n| \cdot \mu_n \cdot \frac{V_{DS}}{L} \cdot w$

$I_n = V_{DS} \cdot g_d$ $\quad g_d = |Q_n| \cdot \frac{\mu_n \cdot w}{L} = C_{ox}(V_G - V_{th}^*) \cdot \frac{\mu_n \cdot w}{L}$

V_{DS} is small, $V_G > V_{th}^* \Rightarrow I_n \propto V_{DS} \propto (V_G - V_{th}^*)$

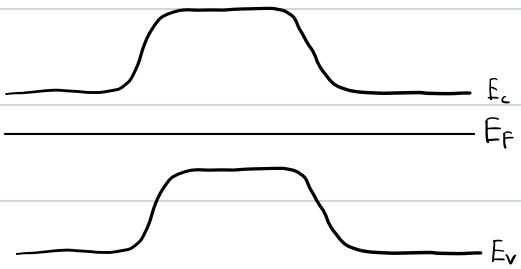


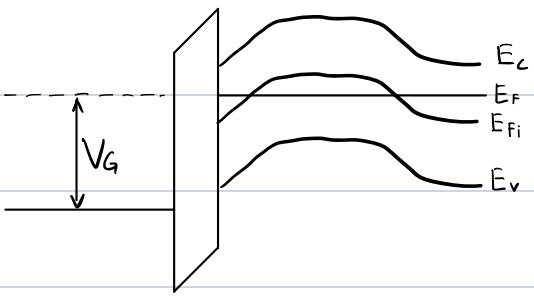


Gate → Body → Source

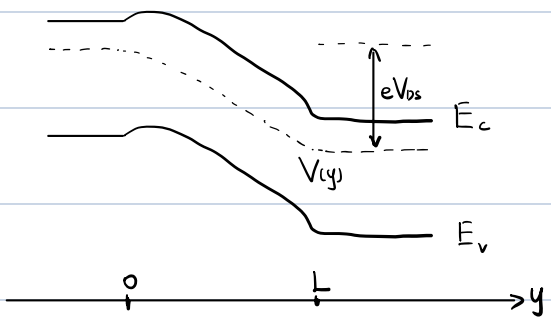
source

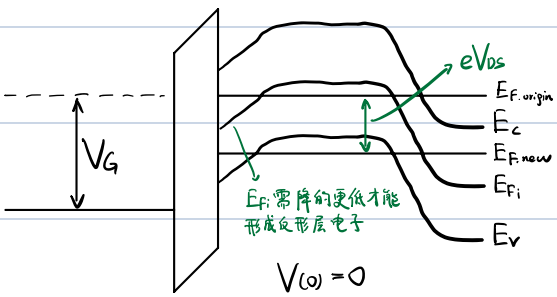
$V_{DS} = 0$





$V_{DS} > 0$





$V(0) = 0$

$V(L) = V_{DS}$

$V_{Flat} = 0$. $V_{th} = \varphi_s + \frac{Q_{ss}}{C_{ox}}$. $\varphi_s = 2\varphi_F$. $\Rightarrow \varphi_s' = 2\varphi_F + V_{DS}$

$$V_{th} = 2\varphi_F + \frac{\sqrt{2\epsilon_s \cdot 2\varphi_F \cdot eN_A}}{C_{ox}}$$

$$\Rightarrow V_{th}' = (2\varphi_F + V_{DS}) + \frac{\sqrt{2\epsilon_s (2\varphi_F + V_{DS}) eN_A}}{C_{ox}}$$

$$\Delta V_{th} = V_{th}' - V_{th} = V_{DS} + \frac{\sqrt{2\epsilon_s (2\varphi_F + V_{DS}) eN_A} - \sqrt{2\epsilon_s 2\varphi_F eN_A}}{C_{ox}}$$

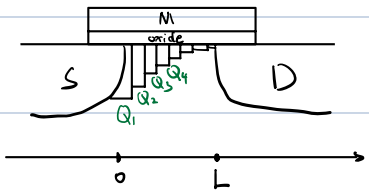
$$|Q_n| = C_{ox}(V_G - V_{th}') = C_{ox}(V_G - V_{th} - \Delta V_{th})$$

$$|Q_n| = C_{ox}(V_G - V_{th} - V_{DS}) - \left(\sqrt{2\epsilon_s (2\varphi_F + V_{DS}) eN_A} - \sqrt{2\epsilon_s 2\varphi_F eN_A}\right)$$

$$\Rightarrow |Q_n(y)| = C_{ox}(V_G - V_{th} - V_{(y)}) - \left(\sqrt{2\epsilon_s (2\varphi_F + V_{(y)}) eN_A} - \sqrt{2\epsilon_s 2\varphi_F eN_A}\right)$$

$V_G > V_{th}^*$. 沟道从 source 到 drain 形成反形层的电荷量.

簡化 $|Q_n(y)| \approx C_{ox}(V_G - V_{th} - V_{(y)})$



$V_{(y)}|_{y=L} = V_{DS}$, $V_{(y)}|_{y=0} = 0$

$I_1 = -W \cdot Q_1 \cdot \mu \cdot E_1$ $\quad Q_1 > Q_2 > Q_3 > Q_4$

$I_2 = -W \cdot Q_2 \cdot \mu \cdot E_2$ $\quad I_1 = I_2 = I_3 = I_4$

$I_3 = -W \cdot Q_3 \cdot \mu \cdot E_3$ $\quad E_1 < E_2 < E_3 < E_4$

$\vdots$

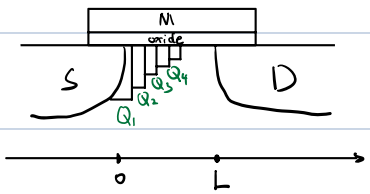
$$I_{(y)} = W \cdot Q_{n(y)} \cdot \mu \cdot \frac{\partial V}{\partial y}$$

$$I_{(y)} = \mu \cdot W \cdot C_{ox} \cdot (V_G - V_{th} - V_{(y)}) \cdot \frac{\partial V}{\partial y} = \text{Constant} = I_{DS}$$

$$I_{DS} = \frac{1}{L} \cdot \int_0^L I_{(y)}\, dy = \frac{\mu \cdot W \cdot C_{ox}}{L} \int_0^L (V_G - V_{th} - V_{(y)}) \cdot \frac{\partial V}{\partial y}\, dy = \frac{\mu \cdot W \cdot C_{ox}}{L} \int_0^{V_{DS}} (V_G - V_{th} - V_{(y)})\, dy$$

$$\Rightarrow I_{DS} = \frac{\mu \cdot W \cdot C_{ox}}{L} \cdot \left[(V_G - V_{th}) \cdot V_{DS} - \frac{V_{DS}^2}{2} \right]$$

前提条件：$V_{DS} \le V_G - V_{th}$



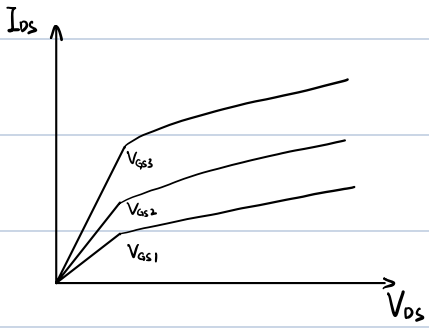
$$v_n = \frac{\mu \cdot E}{1 + \left| \frac{E}{E_c} \right|} \qquad E_c \sim 10^4\ V/cm$$

short - channel device $\longrightarrow I_{DS,sat} = v_{sat} \cdot W \cdot C_{ox} (V_G - V_{th}) \propto (V_G - V_{th})$

when $V_{DS} > V_G - V_{th}$, pinch - off voltage

long - channel device $\longrightarrow I_{DS,sat} \propto \frac{(V_G - V_{th})^2}{2}$

$$\text{Transconductance}: g_m = \left. \frac{\partial I_{DS}}{\partial V_{GS}} \right|_{V_{DS}} = \begin{cases} \frac{\mu \cdot W \cdot C_{ox}}{L} (V_{GS} - V_{th}) & \text{long} \\ v_{sat} \cdot W \cdot C_{ox} & \text{short} \end{cases}$$

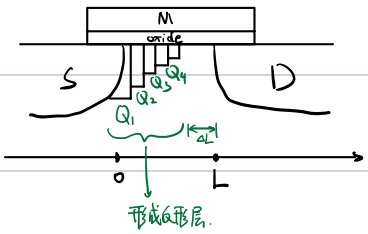


$$I_{DS} = \frac{1}{L-\Delta L}\int_0^{L-\Delta L} I_{(y)}\,dy = \frac{\mu\cdot W\cdot C_{ox}}{L-\Delta L}\int_0^{L-\Delta L}(V_G - V_{th} - V_{(y)})\cdot\frac{\partial V}{\partial y}\,dy = \frac{\mu\cdot W\cdot C_{ox}}{L-\Delta L}\int_0^{V_{GS}-V_{th}}(V_G - V_{th} - V_{(y)})\,dy$$

$$\Rightarrow I_{DS} = \frac{\mu\cdot W\cdot C_{ox}}{L-\Delta L}\cdot\frac{(V_{GS}-V_{th})^2}{2}$$

$V_{DS}\uparrow \Rightarrow \Delta L\uparrow \cdot\ I_{DS}\uparrow$

$$\Rightarrow I_{DS} = \mu\cdot W\cdot C_{ox}\cdot\frac{(V_{GS}-V_{th})^2}{2}\cdot(1+\lambda\cdot V_{DS})$$



output resistance : $r_{out} = \frac{\partial V_{DS}}{\partial I_{DS}} = \left[\mu W\cdot C_{ox}\cdot\frac{(V_{GS}-V_{th})^2}{2}\cdot\lambda\right]^{-1}$